



JOINT VALIDATION AND VERIFICATION OF “SPECTRA SOLAR PARK LTD _35 MW AC GRID-TIED BY SPECTRA SOLAR PARK LIMITED”



Document Prepared by LGAI Technological Center S.A.
(Applus+ Certification)

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Summary:

Validation purpose: The main purpose of this project activity is to reduce GHG emission by installing Solar Photovoltaic (SPV) Panels by Spectra Solar Park Limited. The 35 MW_{AC} grid tied solar power project comprising of Solar PV module (Poly-crystalline), grid tied inverters, power transformers, power evacuation infrastructure, energy metering points, switch yards and other civil constructions is implemented at a site where no other power plant was operating prior to the implementation of the project activity (green-field power plant).

Spectra Solar Park Limited (hereafter referred as “project proponent”) is installing the complete solar power plant (hereafter referred as “proposed project activity”) at located at Nali in Shibalo, Upazila of Manikganj District under Dhaka Division of Bangladesh. Spectra Solar Park Limited is looking forward to earn carbon credits under VCS mechanism for the installed solar power project.

The details of the project activity are as follows:

S.I No.	Location	Installed Capacity (MW _{AC})	Date of Commissioning
1	Nali in Shibalo, Upazila of Manikganj District under Dhaka Division of Bangladesh.	35	12-March-2021

During the 7 years of first crediting period i.e., 12-March-2021 to 11-March-2028, the project will replace anthropogenic emissions of greenhouse gases (GHG's) estimated to be approximately 37,042 tCO₂e per year, thereon total GHG emission reductions for the chosen 07-year crediting period will be 259,292 tonnes of CO₂e.

The objective of this validation activity is to have an independent third party for the assessment of the project design, estimated ER sheet and to ensure a thorough assessment of the proposed project activity against the applicable CDM and VCS requirements. In particular;

- The project's monitoring plan is assessed against “ACM0002 Grid-connected electricity generation from renewable sources --- Version 21.0”/9/.
- The projects compliance with the requirements of Article 12 of the Kyoto Protocol, the CDM Modalities and Procedures as agreed in the Marrakech Accords under decision 3/CMP.1, the annexes to this decision, subsequent decisions and guidance made by COP/MOP & CDM Executive Board and other relevant rules, including the Host Country legislation and sustainability criteria along with VCS guideline and standard version
- CDM Validation and Verification Standard for project activities, version 03.0/6/
- VCS standard, v4.4/9/
- VCS Program Guide v4.3/9/

Validation is a requirement for all VCS projects and is seen as necessary to provide assurance to stakeholders of the quality of the project and its intended generation of estimated verified carbon units (VCUs).

A risk-based approach has been followed to perform this joint validation and verification activity. In the course of Joint Validation & Verification, 09 Corrective Action requests (CARs) and 07 Clarification Requests (CLs), 01 Forward action request (FARs) were raised and successfully closed (for joint validation & verification). The review of the project description and additional documents related to baseline and monitoring methodology; the subsequent background investigation, follow-up interviews

and project owners have provided LGAI Technological Center S.A. (Applus+ Certification) with sufficient evidence to verify the fulfilment of the stated criteria of VCS

Verification purpose:

The main purpose of this project activity is to reduce GHG emission by generating clean energy (electricity) by harnessing renewable energy (solar) resources and curbing emission of greenhouse gas, the proposed project activity by Spectra Solar Park Limited installing Solar Photovoltaic Panels in Manikganj district of dhaka division of Bangladesh by Spectra Solar Park Limited.

Spectra Solar Park Limited (hereafter referred as “project proponent”) is installing solar photo voltaic panels (hereafter referred as “proposed project activity”) at Nali in Shibaloy, Upazila of Manikganj District under Dhaka Division of Bangladesh. Spectra Solar Park Limited is looking forward to earn carbon credits under VCS mechanism for these SPVs installed under the project.

During the Current Monitoring Period from 12-March-2021 to 31-January-2022 (including first and last dates) the project activity contributed to the GHG reductions 33,223 tCO_{2e}.

During the current monitoring period, project activity undergoes continued operation since their commissioning and no major breakdown had taken place.

The objective of this verification activity is to have an independent third party for the assessment of the project design, Actual ER sheet and to ensure a thorough assessment of the proposed project activity against the applicable CDM and VCS requirements. In particular;

- The project’s baseline is assessed against “ACM0002 Grid-connected electricity generation from renewable sources --- Version 21.0”/9/.
- The project’s monitoring plan is assessed against “ACM0002 Grid-connected electricity generation from renewable sources --- Version 21.0”/9/.
- The projects compliance with the requirements of Article 12 of the Kyoto Protocol, the CDM Modalities and Procedures as agreed in the Marrakech Accords under decision 3/CMP.1, the annexes to this decision, subsequent decisions and guidance made by COP/MOP & CDM Executive Board and other relevant rules, including the Host Country legislation and sustainability criteria along with VCS guideline and standard.
- CDM Validation and Verification Standard for project activities, version 03.0/6/
- VCS standard, v4.4/9/
- VCS Program Guide,v4.3

A risk-based approach has been followed to perform this joint validation and verification activity. In the course of validation + verification, 09 Corrective Action requests (CARs) and 07 Clarification Requests (CLs), 01 Forward action request (FARs) were raised and successfully closed. The review of the Monitoring report and additional documents related to baseline and monitoring methodology; the subsequent background investigation, follow-up interviews and project owners have provided LGAI Technological Center S.A. (Applus+ Certification) with sufficient evidence to verify the fulfillment of the stated criteria of VCS.

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1 INTRODUCTION

1.1 Objective

LGAI Technological Center S.A. (Hereinafter referred as Applus + Certification) has been appointed by “**Spectra Solar Park Limited**” to perform the Joint validation and verification of the project entitled “Spectra Solar Park Ltd _35 MW AC Grid-Tied by Spectra Solar Park Limited”, VCS standard V 4.4 and VCS Program Guide V4.3. The objective of this joint validation & verification activity is to have an independent third party for the assessment of the project design, ER sheet and to ensure a thorough assessment of the proposed project activity against the applicable CDM and VCS requirements. In particular;

- The project’s baseline is assessed against “ACM0002 Grid-connected electricity generation from renewable sources --- Version 21.0”/09/
- The project’s monitoring plan is assessed against “ACM0002 Grid-connected electricity generation from renewable sources --- Version 21.0”.
- The projects compliance with the requirements of Article 12 of the Kyoto Protocol/06/, the CDM Modalities and Procedures as agreed in the Marrakech Accords under decision 3/CMP.1, the annexes to this decision, subsequent decisions and guidance made by COP/MOP & CDM Executive Board and other relevant rules, including the Host Country legislation and sustainability criteria along with VCS guideline V4.2 and standard version V4.3/09/.
- CDM Validation and Verification Standard for project activities version 03.0/06/
- VCS standard, v4.4/9/
- VCS Program Guide, v4.3

Joint Validation & verification is a requirement for all VCS projects and is seen as necessary to provide assurance to stakeholders of the quality of the project and its intended generation of estimated verified certified units (VCUs). This report contains the findings and resolutions from the validation and verification of the project activity.

1.2 Scope and Criteria

The scope of the Joint validation and verification is the independent and objective review of the Joint Project Description & Monitoring Report. The Joint VCS PD & MR/04/ are reviewed against the relevant criteria (see 1.1) and decisions by the CDM Executive Board and VCS executive board, including the approved baseline and monitoring methodology. The validation and verification were based on the guidance given in the VCS Program Guide, V4.3 and VCS standard, v4.4.

The assessment team has employed a risk-based approach to assess the completeness and accuracy of the claims and conservativeness of the assumptions in the Joint VCS PD & MR. The main focus of the assessment team is to identify the significant risks for the project implementation and the generation of VCUs. The validation and verification are not meant to provide any consulting towards the project participants. However, stated requests for clarifications and/or corrective actions may have provided input for improvement of the project design and monitoring report combined.

The only purpose of the validation and verification is its usage during the registration /issuance process as part of the VCS project cycle. Therefore, LGAI Technological Center S.A. (Applus+ Certification) can't be held liable by any party for decisions made or not made based on the validation/verification opinion, which will go beyond that purpose.

1.3 Level of Assurance

The verification and validation have been planned and organized to achieve a Reasonable Level of assurance as per the requirement of VCS.

1.4 Summary Description of the Project

The main purpose of this project activity is to reduce GHG emission by installing Solar Photovoltaic (SPV) Panels and generating clean energy (electricity) by harnessing renewable energy (solar) resources and curbing emission of greenhouse gas Projects in Manikganj District of Dhaka region of Bangladesh by Spectra Solar Park Limited. Electricity generated from the project activity is supplied to Bangladesh Power Development Board (BPDB), thus substitutes high carbon intensive grid electricity by clean and renewable based electricity

The project was commissioned on 12-March-2021. The commissioning certificate /01/ and power purchase agreement /14/ is reviewed by the verification team and found to be appropriate. The verification team has also arranged co-call with site personnel and verified the latest photographs /15/ of the project equipment's and monitoring arrangements. The project activity instance was found to be implemented as it has been defined in the joint VCS PD & MR.

The details of project capacity and location details for all the project instances are as follows: -

S.I. No.	Location	Installed Capacity (MW _{AC})	Date of Commissioning
1	Nali in Shibaloy, Upazila of Manikganj District under Dhaka Division of Bangladesh.	35	12-March-2021

During the 7 years of first crediting period i.e., 12-March-2021 to 11-March-2028, the project will replace anthropogenic emissions of greenhouse gases (GHG's) estimated to be approximately 37,042 tCO_{2e} per year, thereon total GHG emission reductions for the chosen 07-year crediting period will be 259,292 tonnes of CO_{2e}.

During the Current Monitoring Period from 12-March-2021 to 31-Jan-2022 (First and last date included) the project activity has contributed 33,223 tCO_{2e} GHG reductions.

2 VALIDATION AND VERIFICATION PROCESS

2.1 Method and Criteria

Validation and Verification Scope: The scope is defined as an independent and objective review of the Joint project design document and Monitoring report. The Joint VCS PD and MR is reviewed against the criteria stated in Article 12 of the Kyoto Protocol/06/, the CDM modalities and procedures as agreed in the Marrakech Accords and the relevant decisions by the CDM Executive Board and VCS standard version 4.4 and VCS Program Guide,V4.3/09/, including the approved baseline and monitoring methodology “ACM0002 Grid-connected electricity generation from renewable sources — Version 21.0”/9/. The validation and verification were based on the requirements in the Validation and Verification Standard for project activities, version 03.0, /06/ and VCS Program Guidev4.3 and VCS standard version 4.3./09/.

The validation and verification are not meant to provide any consulting towards the project participants. However, stated requests for clarifications and/or corrective actions may have provided input for improvement of the combined project document and the Monitoring report.

Validation and Verification Process: The project assessment is based on the Clean Development Mechanism Validation and Verification Standard for project activities version 03.0 and VCS standard v4.3 and Program Guide v4.3 and is conducted using standard auditing techniques to assess the correctness of the information provided by the project participants. Before the assessment begins, members of the team covering the technical scope(s), sectoral scope(s), and relevant host country experience for evaluating the VCS project activity are appointed.

Once the project is received by the assessment team, the members of the assessment team carried out:

- A desk review of the Joint project design documentation and monitoring report;
- Follow-up interviews with project participant;
- The resolution of outstanding issues and the issuance of the final Joint –VAL & Verification report and opinion.

In order to ensure transparency, assumptions must be clear and stated explicitly and background material must also be referenced. LGAI Technological Center S.A. (Applus+ Certification) has developed a specific checklist customized for the project. The checklist demonstrates, in a transparent manner, the project criteria (requirements), discussion on each criterion by the assessment team, and the results from validating/verifying the identified criteria.

Appointment of the assessment team: -

According to the sectoral scope / technical area and experience in the sectoral or national business environment, LGAI Technological Center S.A. (Applus+ Certification) has composed a project assessment team in accordance with the appointment rules in the internal Quality Management System of LGAI Technological Center S.A. (Applus+ Certification).

The composition of audit team shall be approved by the LGAI Technological Center S.A. (Applus+ Certification) ensuring that the required skills are covered by the team.

The four qualification levels for team members that are assigned by formal appointment rules are as presented below:

- Lead Auditor (LA).
- Auditor (A) / Auditor in Training (AiT).
- Technical Expert (TE).
- Technical Reviewer (TR).

The sectoral scope / technical area knowledge linked to the applied methodology/ies shall be covered by the assessment team.

Name	Role	SS Coverage	TA Coverage	Financial aspect	Host country experience
Jitendra Mohan Singh	LA/TE	YES	YES	YES	YES
Denny Xue	TR	YES	YES	YES	NA

The complete list of CVs is included as Appendix 3 of this report.

Document Review: -

The Joint VCS PD & MR submitted by the Client was reviewed against the approved methodology and other relevant criteria to verify the correctness, credibility, and interpretation of the presented information. Furthermore, a cross-check between information provided and information from other sources has been done. A complete list of all documents and evidence material reviewed is included in this report below in appendix 1.

Follow-up interview

A remote audit & interviews are conducted by LGAI Technological Center S.A. (Applus+ Certification) assessment team who performed the teams interview of site personnels, inspection of each project site with project stakeholders to confirm selected information and to resolve issues identified in the document review. The detail is provided in this report in the below sections.

Resolution of Clarification and Correction Action Request: -

The objective of this phase of the joint validation and verification was to resolve the requests for corrective actions and clarification and any other outstanding issues which need to be clarified for LGAI Technological Center S.A. (Applus+ Certification) positive conclusion on the project design and Monitoring report. The Corrective Action Requests and Clarification Requests raised by LGAI Technological Center S.A. (Applus+ Certification) were

resolved during communications between the Client and Applus+ Certifications to guarantee the transparency of the validation process, the concerns raised and responses given are summarized below in the appendix 2.

The Joint VCS PD & MR Version 04/04/ submitted by project owners on 24-February-2023 serves as the basis for the final assessment presented. Additional changes to the project during the joint validation and verification process are not considered to be significant with respect to the main CDM/VCS objectives. The two CDM/VCS main objectives are the reduction of anthropogenic GHG emissions and the contribution of sustainable development to the host country.

Internal quality Control: -

As final step of a joint validation and verification of the final documentation including the final Joint validation and verification report and the checklist have to undergo an internal quality control by the technical review committee, i.e., each report has to be finally approved either by the head of the technical review committee or the deputy. In case one of these two persons is part of the assessment team approval can only be given by the other one to avoid any conflict of Interest.

After confirmation of the project owners, the positive validation/verification opinion and relevant documents are submitted to the VCS secretariat through the VCS web-platform.

2.2 Document Review

The details of the document observed during the joint validation and verification process are listed below in appendix 1 of this report.

2.3 Interviews

The audit was conducted through video conferencing on dated 12-December-2022 based on the Risk Analysis for conducting Remote audit/18/. In accordance with the paragraph 4.1.9 of VCS standard v.4.4, Site visit shall be conducted under the following circumstances:

- 1) The first verification of the project after validation;
- 2) Verification of project baseline reassessments; and,
- 3) Verifications that assess a project description deviation where the deviation impacts the applicability of the methodology, additionality or the appropriateness of the baseline scenario.

This Joint Validation and Verification of the PA. VVB has conducted risk analysis of conducting remote audit/18/ before interview which is in line with the requirement of the paragraph 4.1.10 of VCS Standard v.4.4. The VVB has taken alternative measures to reach reasonable level of assurance and conducted remote audit using ICT tool “zoho meeting”/Telephone with site personal & consultant (refer section 2.3) with the PP representative.

Technical details & metering/monitoring arrangement verified through onsite photographs/name plates/15/ and calibration certificates/10/ shared by PP. All the documents were cross checked to ensure conservative estimation of emission reduction. During the remote audit, the PP representatives were questioned about the implementation of the project activity. Several topics

like commissioning date, the generation, recording, and monitoring of the data and the error accountability were discussed. To cross check the information provided by PP, various documents like technical specifications, commissioning certificates/1/, Power Purchase Agreement PPA/14/, JMR sheets/11/, invoice/11/, calibration certificate/10/ etc. were also verified. Moreover, FAR 01 has been raised for next verifying VVB.

The names of the persons/local stakeholders interviewed during remote audit through Zoho meeting on 12-December-2022 is given below:

Sr. No.	Interviewee			Subject	Team member
	Last name	First name	Affiliation		
1.	Huda	Mr. Kamrul	Project Director	Project Implementation, JMR & invoicing procedure, calibration, grievance mechanism, Management practices, data storage, QA/QC	Mr. Jitendra Mohan Singh (Team Leader/Technical Expert/Financial Expert)
2.	-	Mr. Moni Rozzaman	Plant Manager		
3.	Majumdar	Surva	Consultant EKI Energy Services Ltd.		
4.	Ali	Haraz	Local stakeholder Village:Nali	Information about project, Local stakeholder meeting, Grievances (if any)	Mr. Jitendra Mohan Singh (Team Leader/Technical Expert/Financial Expert)
5.	Miah	Motelab	Local Stakeholder Village: Nayakandi		
6.	Munsi	Pashan	Local Stakeholder Village:Nali		

2.4 Site Inspections

Duration of remote audit: 12-December-2022				
No.	Activity performed on-site	Site location	Date	Team member
1.	Assessment team checked the implementation of the project, Baseline emission, Emission reduction calculation, technical description of the project and Monitoring.	Interview through video conferencing using zoho meeting ICT tool	12- Decemeber- 2022	Mr. Jitendra Mohan Singh (Team Leader/Technical Expert/Financial Expert)

2.5 Resolution of Findings

The objective of this phase of the joint validation and verification was to resolve the requests for corrective actions and clarification and any other outstanding issues which need to be clarified for LGAI Technological Center S.A. (Applus+ Certification)'s positive conclusion on the project design and Monitoring report. The Corrective Action Requests and Clarification Requests raised by LGAI Technological Center S.A. (Applus+ Certification) were resolved during communications between the Client and LGAI Technological Center S.A. (Applus+ Certification) to guarantee the transparency of the validation process, the concerns raised and responses given are summarized below in the Appendix 2.

The final Joint VCS PD & MR Version 04/04/ submitted by project owners on 24-February-2023 serves as the basis for the final assessment presented. Additional changes to the project during the validation and verification process are not considered to be significant with respect to the main CDM/VCS objectives. The two CDM/VCS main objectives are the reduction of anthropogenic GHG emissions and the contribution of sustainable development to the host country.

Areas of validation and verification findings	No. of CL	No. of CAR	No. of FAR
Project design document and Monitoring report	00	01	00
Description of project activity	04	02	01
Application of selected baseline and monitoring methodology and selected			
- Applicability of methodology and standardized baseline	00	00	00
- Deviation from methodology	00	00	00
- Clarification on applicability of methodology, tool and/or standardized baseline	01	01	00
- Demonstration of additionality	01	01	00
- Emission reductions	01	01	00
- Monitoring plan	00	01	00
- Stakeholder's consultation process	00	01	00
- Public comments	00	00	00
Others (please specify)-Matter related to double counting- for validation	00	00	00
Others (please specify)-	00	00	00

Areas of validation and verification findings	No. of CL	No. of CAR	No. of FAR
ER achieved – Actual ER achieved calculations, Compliance of the registered monitoring plan with the methodologies including applicable tools and standardized baselines Compliance of monitoring activities with the registered monitoring plan Compliance with the calibration frequency requirements for measuring instruments Assessment of data and calculation of emission reductions or net removals Assessment of reported sustainable	00	01	00
Total (Validation+ Verification)	07	09	01

The list of findings and their resolution is presented in Appendix 2 of this report.

2.5.1 Forward Action Requests

One FAR was raised during this joint validation and verification process. Please refer Appendix 2 for details.

3 VALIDATION FINDINGS

3.1 Project Details

The main purpose of this project activity is to reduce GHG emission by installing Solar Photovoltaic (SPV) Panels and generating clean energy (electricity) by harnessing renewable energy (solar) resources and curbing emission of greenhouse gas Projects in Manikganj District of Dhaka region of Bangladesh by Spectra Solar Park Limited. Electricity generated from the project activity is supplied to Bangladesh Power Development Board (BPDB), thus substitutes high carbon intensive grid electricity by clean and renewable based electricity

Baseline Scenario: -

The project intends to use the electricity generated from the solar power activity supplied to Bangladesh Power Development Board (BPDB), thus substitutes high carbon intensive grid electricity by clean and renewable based electricity. The solar plant unit is located at the Manikganj district of Dhaka region of Bangladesh. The project activity displacing electricity from Bangladesh Grid, which is supplied by predominantly fossil fuel plants. The project activity instances would replace equivalent quantum of fossil fuel dominated grid power that's being used prior to the project activity instances. Hence, pre-project scenario and baseline scenario are the same.

Estimated Average GHG emission reductions from the project activity will be 37,042 tonnes of CO₂e per year and total GHG emission reductions for the chosen 7-year crediting period will be 259,292 tonnes of CO₂e. This is a project activity and geographical boundary of the project activity is fixed as Bangladesh only. Hence, all the project instances shall be located within the Bangladesh only.

During the Current Monitoring Period from 12-March-2021 to 31-January-2022 (First and last date included) the project activity has contributed 33,223 tCO₂e GHG reductions.

The operational life of the installed solar module is 25 years based upon the standard operational and maintenance practices followed at site. Same is also verified from the PPA, Operation & Maintenance Agreement/14/ and technical details of the project activity.

The technical specifications/07/ of the equipment's installed for the project activity are as follows:-

Sr	Description of Items		Key Technical Specification
1	Connected Capacity		35 MW _{AC}
2	Solar PV Module	Make	SUNTECH
		PV module technology	Poly Crystalline Silicon
		Unit Nom. Power	330 Wp
		No of Module	Total number of PV modules - 137,520 of 330 Wp capacity
3	Grid tied Inverter	Make	SUNGROW
		Inverter technology	Central inverters
		No of Inverter	12
		Rated Output Voltage	AC 600 V, 3~
		Frequency range	50Hz±5%
		Max. Apparent Power	3593 kVA
		Rated output power	3125 kVA
4		Transformer	Number of Transformer - 6 Nos. Capacity – 6.25 MVA Output Voltage – 33kV
5		Collector Plane Orientation	14 ^o
6	Energy Meter (Main Meter)		Make - Landis+Gyr E650 Type – Three Phase 4 wire Accuracy Class- 0.2S SL. No. 42334949 Location of the meter- Solar Power plant of Spectra
8	Energy Meter (Check meter)		Make - Landis+Gyr E650 Type – Three Phase 4 wire Accuracy Class- 0.2S SL. No. 42334947 Location of the meter- Solar Power plant of Spectra
8	Substation details	Power plant is connected to the nearest 33/11 substation named PATURIA 33/11 KV PBS Substation	

Assessment team checked during the remote audit interview and confirms that the details of the project proponent areas below

Organization name	Spectra Solar Park Limited
Contact person	Khan Md. Aftabuddin
Title	Managing Director
Address	House: 17, Road: 106, Block: CEN(F), Gulshan-2, Dhaka-1212, Bangladesh
Telephone	880-255059854, 880-2-58816192
Email	chairmanspectragroup@gmail.com

Other entity involved is EKI Energy as project consultant and details are as below:

Organization name	EKI Energy Services Ltd.
Role in the project	Project Consultant
Contact person	Mr. Surva Majumdar
Title	General Manager, Operations
Address	Office no. 201, Plot 48, Scheme 78 Part 2 Vijay Nagar, Near Brilliant Convention Centre Indore - 452010 (M.P, India) Website www.enkingint.org
Telephone	+91-8989880655
Email	suvra.majumdar@enking.org / registry@enkingint.org

Project start Date: -

Project Start Date: 12-March-2021 which is the date of commissioning of the project activity.

Assessment team checked the commissioning details from the commissioning certificates/01/ provided by the project proponent and start date found correct.

Project crediting period: -

Assessment team confirms that the crediting period dates for the project is as below: -

Project crediting period: Renewal crediting period

Start date of Crediting period: 12-March-2021

End date of crediting period: 11-March-2028

Total number of years: 07

The project activity adopts renewable crediting period of 07 years which can be renewed for maximum 2 times.

Project scale & estimated GHG Emissions Reductions or Removals: -

Since the installed capacity of the project activity is 35 MW_{AC}. Assessment team confirms that the scale of the project activity as per the section 3.10.1 of VCS standard version 4.4,/09/ the projects are classified as follows:

- 1) Projects: Less than or equal to 300,000 tonnes of CO₂e per year
- 2) Large Projects: Greater than 300,000 tonnes of CO₂e per year

The project activity instances being included currently have less than 300,000 tCO₂e Emission reductions, hence these project activity instances are classified as "Projects"

Project Scale	
Project	✓
Large project	

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
Year 1	37,827
Year 2	37,562
Year 3	37,299
Year 4	37,038
Year 5	36,779
Year 6	36,521
Year 7	36,266
Total estimated ERs	259,292

Total number of crediting years	7
Average annual ERs	37,042

Condition prior to project initiation: -

Assessment team during the desk review and onsite assessment confirms that the project is an installation of solar photovoltaic by Spectra Solar Park Limited and does not involve generation of GHG emissions for the purpose of their subsequent reduction, removal or destruction. The baseline as described in section 3.4.4 of this report will continue to be the baseline in the absence of project activity.

Project compliance with applicable laws, statutes, and other regulatory framework: -

Assessment team confirms that the Project has received necessary approvals for development and commissioning for the proposed Solar PV project from the government of Bangladesh and is in compliance to the local laws and regulations. Assessment team checked the Commissioning certificates, power purchase agreements, Installation report for Solar Plants to confirm the project capacity and its relevant statutory requirements as per the host country regulations.

The project is a voluntary initiative by the project proponent and has not been implemented to meet any local/national laws or regulatory compliances /09/. The project activity is in compliance with current laws and regulations and there are no legal and / or regulatory requirements that prevent the project implementation. Also, the assessment team has confirmed that there is no such compliance requirement with an emission trading program or any binding limits on GHG emissions for the project activity in Bangladesh (host country) as it is a Least Developed Country (LDC). The project has obtained valid consents, installation and operation from the state nodal agencies and is in compliance with local laws and regulations.

Project ownership: -

Spectra Solar Park Limited is project proponent (PP) of project activity and they have the legal right to control and operate the project activities.

The project ownership has been checked by the Assessment Team and demonstrated through below supporting documents:

- Power Purchase Agreement (PPA)/ 14/
- Commissioning certificate/01/

Further, individual compliance aspects will be provided for each of the project activity instance to be included in the project activity based on the clearances/09/certificate received for implementation of project instance.

Emission Trading programs & other binding limits: -

Assessment team confirms that the Net GHG emission reductions or removals generated by the Project will not be used for compliance with an emissions trading program or to meet

binding limits on GHG emissions in any Emission Trading program or other binding limits. Further, Declaration in effect of the same has been submitted by project proponent to audit team and found to be correct. Thus, it is concluded that the project activity not involved on other Emissions trading programs and other binding limits.

Leakage Management for AFOLU project: -

Not applicable to the project activity.

Commercially Sensitive Information: -

No commercially sensitive information has been excluded from the public version of the project description. The details are presented transparently to the assessment team for analysis which lead to positive conclusion for this validation and verification.

Sustainable Development: -

Sustainability is defined by three different parameters : environmental sustainability, social sustainability, and economic sustainability.

- The project activity utilizes renewable energy source for electricity generation instead of fossil fuel-based electricity generation which would have gaseous, liquid and/or solid effluents /wastes. Thus, the project causes no negative impact on the surrounding environment and contributes to environmental well being.
- The project activity supporting to alleviate poverty by means of direct and indirect employment generation and improves activities by strengthening of local grid of the state electricity utility. Thus, contributes livelihood to the society.
- The project generates electricity utilizing renewable source i.e. solar. Thus, it will increase the contribution of renewable based power generation in the region and will also help in reducing the demand – supply gap of the respective grid.
- The project activity involves substantial amount of investment, manpower, operational amenities etc. Thus, it contributes to direct and indirect employment generation as per the requirement of skilled and semi skilled manpower, both during construction and operational phase.

Further, inline with VCS Standard Version 4.4/09/, “The project proponent must demonstrate that a project contributes to at least three SDGs by the end of the first monitoring period, and in each subsequent monitoring period.” As per the provision of implementation deadline, “Projects registered before 20 January 2023 shall demonstrate contributions to at least three SDGs by 20 January 2025”. However, PP have provided the information on the SDG contributions by the project activity for SDGs:

1. SDG 7.2: Renewable energy share in the total final energy consumption.
2. SDG 8.5: Unemployment rate, by sex, age and persons with disabilities
3. SDG 13.0: Tonnes of greenhouse gas emissions avoided or removed.

VVB have assessed the SDG 7.2, SDG 08 and SDG 13.0 and supporting documents shared by PP and found values correct as provided in Section 1.17.2 of Joint VCS PD & MR version 04/04/.

Hence, in view of the assessment of VCS joint PD & MR /04/ and supporting documents are listed in Appendix 1 of this report. The validation team is able to confirm that the description contained in the VCS Joint PD and MR of the project activity and the technical aspects of its implementation. Consequently, assessment team confirms that the project description of the project contained in the joint VCS PD and MR /04/ to be complete and accurate. The VCS joint PD and MR complied with relevant forms and guidance for completing the VCS joint PD and MR.

3.2 Participation under Other GHG Programs

The project has neither been registered nor seeking registration under any other GHG programs. The project is seeking registration only in VCS program. Further, declaration for the same is checked and found correct by the assessment team. Also, assessment team checked the following registries to confirm the same. The details of the registries checked are as follows:

1. <http://cdm.unfccc.int/>
2. <http://www.goldstandard.org/>

Rejection by other GHG program: -

The Project is not rejected by other GHG programs. A declaration for the same is checked and found correct by the assessment team. Also, assessment team checked the following registries to confirm the same. The details of the registries checked are as follows:

1. <http://cdm.unfccc.int/>
2. <http://www.goldstandard.org/>
3. www.verra.org

The Project does not intend to generate any other form of GHG-related environmental credit for GHG emission reductions or removals claimed under the VCS Program.

3.3 Safeguards

3.3.1 No Net Harm

The project activity promotes environmental and socio-economic well-being as it results in zero GHG emissions due to installation and operation of clean, renewable energy technology for electricity generation.

3.3.2 Local Stakeholder Consultation

As per the VCS requirements, it is necessary to invite the relevant stakeholders, prior of the validation process. Moreover, the stakeholder meeting took place prior to the start date of the project activity which fulfill the requirement VCS Standard 4.4. The VVB checked the relevance of the dates during the validation remote interview audit. The detail of the invitation date and stakeholder meeting date is as below:

Sr. No.	Local Stakeholder Meeting Date	Venue
1.	13-February-2019 to 15-February-2019	ESIA consultant with support from representative of project proponent carried out wide stakeholders' consultation at 10 locations involving wider group of stakeholders
2.	14-February-2019	Project proponent carried out stakeholder's consultation meeting at the plant premises involving representative from nearby villages.

The local stakeholders' consultation meeting was attended by local persons including local villagers, local vendors, and technology suppliers.

The stakeholders identified by the project participant were local people/ NGOs who are the major population of the area, local communities, technology supplier, project proponent representatives, O&M Team and other people involved in the project. Validation team verified the list of participants who attended the stakeholder meeting and feedback questionnaire and confirms the stakeholders identified are relevant. The validation team also verified the minutes of meeting to note that no negative comments were received and the same was cross checked with the information obtained during follow up interviews with the stakeholders. Moreover, assessment team during the validation interview also noted that a grievance register is also put onsite for the stakeholder to comment on any grievances during the operation lifetime of the project activity, the grievances from the stakeholder if found suitable will be addressed immediately by the top management and thus the approach is found appropriate for the project activity.

Further, Complaint/suggestion/feedback register/13/ is maintained at site as a part of ongoing communication with stakeholders in line with VCS Standard, ver. 4.4 and appropriate actions taken time to time by PP.

Assessment team checked the grievance register provided by PP and found that local stakeholders can anytime lodge their grievances if any in the register over the operational life time of the project. During current monitoring period no grievance was received. Thus, assessment team is of the opinion that the ongoing stakeholder mechanism is adequate and appropriate.

3.3.3 Environmental Impact

The ESIA assessment for the project has been prepared by the ERM India Private limited /reference/. The report was approved by the Ministry of Environment, Government of Bangladesh/16/. The ESIA report for the project covers all aspects of the project including capacity, interaction with other plants in the vicinity, natural resources of the project including management, social and economic impacts, technology and materials used, current land use

in the region, any historical or protected site within the project boundaries, geological assessment of the project site and any communities affected by the project.

Thus, the adequacy of the analysis carried out by the project participant for the impact of the implementation of the project activity on the environment has been validated as per the VVS standard for project activity version 03.0/06/

3.3.4 Public Comments

Assessment team noted that this project was open for public comment from 13-May-2022 to 12-June-2022. No comments were received. The details were checked by the assessment team in the following web platform¹.

3.3.5 AFOLU-Specific Safeguards

The project activity is not AFOLU project; Hence this section is not required.

3.4 Application of Methodology

3.4.1 Title and Reference

Assessment team checked that following methodology and tools are applicable for the project activity. The details are as below:

Methodology:

Methodology: "ACM0002 Grid-connected electricity generation from renewable sources --- Version 21.0"²

Sectoral scopes: 1

The methodology refers to following CDM tools:

- Tool 07: Tool to calculate the emission factor for an electricity system - Version 07.0³

¹<https://registry.terra.org/app/projectDetail/VCS/3099>

² <https://cdm.unfccc.int/UserManagement/FileStorage/ZPFJL01OU2RYC6N3HASIXV7K84QBG9>

³ <https://cdm.unfccc.int/methodologies/PAMethodologies/tools/am-tool-07-v7.0.pdf>

- Tool 21: Tool 01 - Tool for the demonstration and assessment of additionality – Version 07.0.0, EB 70 annex 08⁴
- Tool 27: “Investment Analysis” Version 8.0⁵ & version 12.0⁵
- Tool 24 – Common practice - Version 03.1, EB 84 annex 7⁶

3.4.2 Applicability

The applied baseline methodology is justified as it has been demonstrated that the proposed project activity is:

Para No.	Applicability Conditions (With para no and reference)	Project Case	VVB Assessment
4	1. This methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Install a Greenfield power plant; (b) Involve a capacity addition to (an) existing plant(s); (c) Involve a retrofit of (an) existing operating plant(s)/unit(s); (d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or (e) Involve a replacement of (an) existing plant(s)/unit(s).	The project activity involves construction and operation of grid-connected solar power (renewable energy) project at a site where no renewable energy power plant was operated prior to the implementation of the project activity and therefore the project activity a “green field power project” as per the definition of the methodology and hence complies to the applicability condition (3.a). Hence the project activity meets the applicability condition of the methodology.	The project activity is greenfield project. Thus, fulfil criteria.
5	2. In case the project activity involves the integration of a BESS, the methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Integrate BESS with a Greenfield power plant; (b) Integrate a BESS together with implementing a capacity addition to	The project activity includes installation of grid connected solar power project without any Battery Energy Storage System (BESS). The electricity generated from the project activity is directly fed to the electricity grid. Therefore, the applicability condition 5 is not applicable/relevant to the project activity.	The project activity is grid connected solar power plant. Hence, No battery is used

⁴ <https://cdm.unfccc.int/methodologies/PAMethodologies/tools/am-tool-01-v7.0.0.pdf>

⁵ <https://cdm.unfccc.int/methodologies/PAMethodologies/tools/am-tool-27-v12.pdf>

⁶ <https://cdm.unfccc.int/methodologies/PAMethodologies/tools/am-tool-24-v1.pdf>

Para No.	Applicability Conditions (With para no and reference)	Project Case	VVB Assessment
	(an) existing solar photovoltaic or wind power plant(s)/unit(s); (c) Integrate a BESS to (an) existing solar photovoltaic or wind power plant(s)/unit(s) without implementing any other changes to the existing plant(s); (d) Integrate a BESS together with implementing a retrofit of (an) existing solar photovoltaic or wind power plant(s)/unit(s).		
6	3. The methodology is applicable under the following conditions: (a) Hydro power plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit; (b) In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects) the existing plant/unit started commercial operation prior to the start of a minimum historical reference period of five years, used for the calculation of baseline emissions and defined in the baseline emission section, and no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity. (c) In case of Greenfield project activities applicable under paragraph 5 (a) above, the project participants shall demonstrate that the BESS was an integral part of the design of the renewable energy project activity (e.g.	The project activity involves construction and operation of greenfield grid-connected solar power project and hence complies to the applicability condition (6.a) of the methodology. Since the project activity does not include capacity additions, retrofits, rehabilitations or replacements of existing plant/unit, the applicability condition 6.b is not applicable/relevant for the project activity. Although the project is a green field project activity but does not include Battery Energy Storage System (BESS). And therefore, the	The current project activity is newly installed solar power not hydro power plant.

Para No.	Applicability Conditions (With para no and reference)	Project Case	VVB Assessment
	by referring to feasibility studies or investment decision documents); (d) The BESS should be charged with electricity generated from the associated renewable energy power plant(s). Only during exigencies may the BESS be charged with electricity from the grid or a fossil fuel electricity generator. In such cases, the corresponding GHG emissions shall be accounted for as project emissions following the requirements under section 5.4.4 below. The charging using the grid or using fossil fuel electricity generator should not amount to more than 2 per cent of the electricity generated by the project renewable energy plant during a monitoring period. During the time periods (e.g. week(s), months(s)) when the BESS consumes more than 2 per cent of the electricity for charging, the project participant shall not be entitled to issuance of the certified emission reductions for the concerned periods of the monitoring period.	applicability condition 6.c is not applicable/relevant for the project activity. The project activity does not include Battery Energy Storage System (BESS). And therefore, the applicability condition 6.d is not applicable/relevant for the project activity.	
7	4. In case of hydro power plants, one of the following conditions shall apply:⁶ (a) The project activity is implemented in existing single or multiple reservoirs, with no change in the volume of any of the reservoirs; or (b) The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density,	The project activity involves construction and operation of greenfield grid-connected solar power project using solar energy for generation of electricity hence the applicability condition “7” is not applicable/relevant to the project activity as the applicability conditions is related to hydro power projects.	Solar power project hence this condition is not applicable.

⁶ Project participants wishing to undertake a hydroelectric project activity that results in a new reservoir or an increase in the volume of an existing reservoir, in particular where reservoirs have no significant vegetative biomass in the catchments area, may request a revision to the approved consolidated methodology.

Para No.	Applicability Conditions (With para no and reference)	Project Case	VVB Assessment
	calculated using equation (7), is greater than 4 W/m²; or (c) The project activity results in new single or multiple reservoirs and the power density, calculated using equation (7), is greater than 4 W/m²; or (d) The project activity is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs, calculated using equation (7), is lower than or equal to 4 W/m², all of the following conditions shall apply: (i) The power density calculated using the total installed capacity of the integrated project, as per equation (8), is greater than 4 W/m²; (ii) Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity; (iii) Installed capacity of the power plant(s) with power density lower than or equal to 4 W/m² shall be: (a) Lower than or equal to 15 MW; and (b) Less than 10 per cent of the total installed capacity of integrated hydro power project.		
8	5. In the case of integrated hydro power projects, project participants shall: (a) Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project; or (b) Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of	The project activity involves construction and operation of greenfield grid-connected solar power project using solar energy for generation of electricity, hence the applicability condition “8” is not applicable/relevant to the project activity as the applicability conditions is related to hydro power projects.	Solar power project hence this condition is not applicable.

Para No.	Applicability Conditions (With para no and reference)	Project Case	VVB Assessment
	reservoirs. The purpose of water balance is to demonstrate the requirement of specific combination of reservoirs constructed under CDM project activity for the optimization of power output. This demonstration has to be carried out in the specific scenario of water availability in different seasons to optimize the water flow at the inlet of power units. Therefore, this water balance will take into account seasonal flows from river, tributaries (if any), and rainfall for minimum of five years prior to the implementation of the CDM project activity.		
9	The methodology is not applicable to: (a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site; (b) Biomass fired power plants/units.	The project activity involves construction and operation of greenfield grid-connected solar power project using solar energy for generation of electricity hence the applicability condition 9" is not relevant as the same pertains to switching from fossil fuels to renewable energy sources or biomass fired power plants/units.	Solar Energy has been utilized for electricity generation, no fossil fuel has been used.
10	In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is "the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance".	The project activity involves construction and operation of greenfield grid-connected solar power project using solar energy for generation of electricity hence the applicability condition 10 is not relevant as the same pertains to retrofits, rehabilitations, replacements, or capacity additions.	The current project activity is newly constructed solar power plant.

LGAI Technological Center S.A. (Applus+ Certification) confirms that the application of the baseline methodology is transparent and conservative and confirms that the chosen baseline and monitoring methodology is applicable to the project activity.

3.4.3 Project Boundary

As per ACM0002, version 21 “The spatial extent of the project extends to the project power plant and all power plant connected physically to the electricity system that the project power plant that connected to through the national grid and to be confined to physical, geographical site of renewable generating units. The boundary also extends to the project power plant and all power plants connected physically to the electricity system that the CDM project power plant is connected to.” Therefore, the entire Bangladesh grid and all connected power plants have been considered in the project boundary for the proposed VCS project activity. The grid is also part of the project boundary as the solar rooftop units are connected to the grid through power evacuation at common point.

The sources and GHG gases involved for proposed Project activity are as below:

Source		Gas	Included?	Justification/Explanation
Baseline	Grid connected electricity generation	CO ₂	Yes	Main emission source
		CH ₄	NO	Minor emission source
		N ₂ O	NO	Minor emission source
Project	Greenfield Solar Power Project activity	CO ₂	NO	No CO ₂ emissions are emitted from the project
		CH ₄	NO	Project activity does not emit CH ₄
		N ₂ O	NO	Project activity does not emit N ₂ O

3.4.4 Baseline Scenario

As per ACM0002, version 21,: “For project activities that displace grid electricity and fossil fuel fired on-site captive electricity, the baseline emission factor should reflect the emissions intensity of the grid and the captive power plant in the baseline scenario i.e. the weighted average emission factor for the displaced electricity is calculated using values based on the historical, prior three year ratios of electricity from captive plants and the grid. For new facilities, the most conservative (lowest) of the emission factor for the two power sources should be used.

Hence, the baseline for the project activity is the equivalent amount of power from the BANGLADESH grid. The combined margin ($EF_{grid,CM,y}$) is the result of a weighted average of two emission factor pertaining to the electricity system: the operating margin (OM) (having weightage 75%) and build margin (BM) (having weightage 25%). Further the values for the operating margin (OM) and build margin and combined margin calculations described in the “Tool to calculate the emission factor for an Electricity System”

The combined margin of the BANGLADESH National Grid used for the project activity is as follows:

Parameter	Value	Source
EF _{grid, CM,y}	0.6145 tCO _{2e} /MWh	Calculated as the weighted average of the operating margin (0.75) & build margin (0.25) values,
EF _{grid, OM,y}	0.6287 tCO _{2e} /MWh	Calculated as the last 3-year weighted average generation and GHG emission for the year 2017-18, 2018-19 and 2019-20 in accordance to Tool to calculate the emission factor for an electricity system (Version 07.0)". The annual value of plant wise generation and operating efficiency is obtained based on Data available from Bangladesh Power Development Board (Annual Report of 2017-18 ,2018-19, 2019-20 ⁷ . Data related to emission factor for each fossil fuel type is obtained from "CDM Baseline of Bangladesh" Department of Environment (DoE)
EF _{grid,BM,y}	0.5722 tCO _{2e} /MWh	Calculated in accordance to Tool to calculate the emission factor for an electricity system (Version 07.0)" based on Data available from Bangladesh Power Development Board (Annual Report 2019-20 and Data related to emission factor for each fossil fuel type is obtained from CDM Baseline of Bangladesh, Department of Environment (DoE)

In the absence of the project activity, equivalent amount of electricity would have been drawn from Bangladesh grid. Hence, the baseline of the project activity is the equivalent amount of power from the national grid.

7

https://bpdh.portal.gov.bd/sites/default/files/files/bpdh.portal.gov.bd/annual_reports/7b792f67_bf50_4b3d_9bef_8f9b568005c9/2022-10-18-05-39-77e3d834e198d21cdfb8d05d5c6c63b0.pdf

7

https://bpdh.portal.gov.bd/sites/default/files/files/bpdh.portal.gov.bd/annual_reports/7b792f67_bf50_4b3d_9bef_8f9b568005c9/2022-10-18-05-38-b38d9a259adcd65ecb411b5ba4637b76.pdf

7

https://bpdh.portal.gov.bd/sites/default/files/files/bpdh.portal.gov.bd/annual_reports/7b792f67_bf50_4b3d_9bef_8f9b568005c9/2022-10-18-05-37-2ae06b4716a042c4b39d5cb54ef20ff4.pdf

Assessment team conclude that the above-mentioned baseline scenario is in line with the applied methodology “ACM0002 Grid-connected electricity generation from renewable sources --Version 21.0”. Same is also reflected in final joint PD and MR version 04. Thus accepted.

3.4.5 Additionality

For demonstrating additionality under VCS, the project activity is required to undergo the following tests

a) **Legal Requirement Test:** - based on the available literature on Electricity Market Law in Bangladesh it was confirmed that there are no enforced laws, statutes, regulations, court orders, environmental-mitigation agreements, permitting conditions or other legally binding mandates requiring its implementation, or requiring the implementation of a similar technology/measure that would achieve equivalent levels of GHG emission reductions.

b) **Additionality Tests:**

As per the applied methodology ACM0002, Version 21.0, additionality of the following project activity is demonstrated and assessed by the latest version of Tool 27, “Investment Analysis” Version 12. The PO has adopted the stepwise approach for demonstrating and assessing the additionality of the project activity as follows:

Project was envisaged for capacity of 35 MW_{AC} in Manikganj of Dhaka region, Bangladesh. Currently, Whole project activity is fully commissioned and continuously contributing towards emission reduction. VCS joint PD and MR for this project activity was open for public comments on 13-May-2022. Start date of the Project is 12-March-2021 which is the commissioning date of whole plants covered under project activity.

The demonstration of additionality for the proposed Project activity is being carried out in accordance with the additionality tool provided by the UNFCCC i.e., Tool 27, “Investment Analysis”, the project proponent shall provide an explanation to show that the project activity would not have occurred anyway due to at least one of the following barriers:

- Investment barrier
- Technological barrier
- Barrier due to prevailing practice
- Other barriers

The project proponent has identified “investment barrier” as the most relevant barrier faced by

the project activity. The investment barrier faced by the project activity consists of barrier due to high capital cost and consequent impact on return.

Investment analysis

Determine whether the proposed project activity is economically or financially less attractive than at least one other alternative. To conduct the investment analysis, use the following sub-steps:

Determine appropriate analysis method and Sub-step 2b (Option III): Apply benchmark analysis

a) Suitability of investment analysis, financial indicator and benchmark:

Project proponent had demonstrated that the financial returns of the proposed VCS project activity would be insufficient to justify the required capital investment as per VCS Standard version 4.4. In the VCS joint PD and MR, project proponent has adopted a conservative approach to identify the benchmark for the project activity. The project is generating revenue in terms of power generated from the Solar power plant being used for selling to the grid. Thus, simple cost analysis (Option I) is not appropriate. Hence out of 2 options, investment comparison analysis (Option II) benchmark analysis (Option III), benchmark analysis is used for the project activity as per project type and decision-making context. Therefore, the Expected return on equity is considered appropriate benchmark. Accordingly, the post-tax Equity IRR has been considered as the relevant financial indicator for the project activity which is acceptable to the assessment team. Moreover, the financial indicator selected by the project proponent is correct based on the fact that tool do not restrict the project proponent to either use project IRR or Equity IRR. This is under the prerogative of the project proponent to select appropriate indicator based on his preferences to know the IRR based on his equity investment or debt investment. The same is thus acceptable to the assessment team. Assessment team however checked the Equity IRR calculation and found that input assumptions used for the calculation of Equity IRR are applicable at the time of investment decision of the project and thus is in accordance with the relevant guideline of the tool.

“In situations where an investment analysis is carried out in nominal terms and the available IRR benchmarks are in real terms, project proponent shall convert the real term values of benchmarks to nominal values by adding the inflation rate. The inflation rate shall be obtained from the inflation forecast of the central bank of the host country for the duration of the crediting period. If this information is not available, the target inflation rate of the central bank shall be used. If this information is also not available, then the average forecasted inflation rate for the host country published by the IMF (International Monetary Fund World Economic Outlook) or the World Bank for the next five years after the start of the project activity shall be used”

The investment analysis has been carried out in Nominal terms. Accordingly, default value has been adjusted by adding suitable forecasted inflation rate taken from ADB (Asian Development Bank, /reference/). Project Participant has calculated Benchmark based on WPI mean inflation rate. The inflation forecast should be for the duration of the crediting period.

As per the Investment analysis tool 27, version 12.0, default value of expected return on equity in real terms for Energy Industries (Group 1) in Bangladesh = 11.73%

VVB observed that the decision to invest in project was done in November 2019 and during this period the applicable version of Investment Analysis was version 10. The default value of expected return on equity was 12.60%⁸. However, at time of Joint Validation & Verification of PA, the investment analysis tool, version 12 is applicable to project activity, in which default value of return of equity is mentioned as 11.73 %. Therefore, 11.73% expected return on equity is considered to be conservative value. Thus, accepted by the assessment team.

Therefore, the Required return on equity (benchmark) was computed in the following manner:

Nominal Benchmark = $\{(1 + \text{Real Benchmark}) * (1 + \text{Inflation rate})\} - 1$

Where:

Default value for Real Benchmark = 11.73% (as per Appendix of CDM TOOL 27)

Inflation Rate forecast for Bangladesh.

Benchmark estimation:

Inflation Rate (at the time of investment decision making)

Inflation forecast as per ADB	Sources Link
5.80	https://www.adb.org/sites/default/files/publication/524596/ado2019-update.pdf (pg 173 of the report) ⁹

Benchmark

Description	Value	Sources Link
Default Value for Bangladesh as per UNFCCC guidelines	11.73%	https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v12.pdf
Inflation forecast	5.80%	https://www.adb.org/sites/default/files/publication/524596/ado2019-update.pdf

⁸ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v10.0.pdf>

⁹ The average forecasted data of Bangladesh for five years was not published by IMF, however calculated average of inflation data for five years after the start of the project activity is around 6.4%. The link of IMF inflation rate is available at the portal <https://www.imf.org/external/datamapper/PCPIPCH@WEO/BGD?zoom=BGD&highlight=BGD>

Benchmark	18.21%	Calculated

b) Parameters and assumptions used:

The project activity is a renewable source of electricity generation and supplies the electricity to the Bangladesh Electricity grid. The key parameters which determine the Equity IRR of the project activity are project cost, PLF and profitability estimates.

In the revised VCS joint PD and MR, the project cost is based on the DPR. The details of the DPR are as below:

Project cost as per the DPR: -

Site Name	Project Proponent	Project Capacity (MW _{AC})	Project Cost (In Million)	DPR Date
Spectra Solar Park Ltd _35 MW AC Grid-Tied	Spectra Solar Park Limited	35	52.064	March 2018

DPR has been submitted to verification team. The DPR was available during decision making and financial profitability of the project was decided based on this DPR. Verification team checked the DPR of the project activity and found that consideration of the project cost in revised VCS joint PD and MR is correct and it is in line with Appendix of Methodological tool 27: "Investment Analysis" Version 12.0 as well as in compliance to VCS Standard. Hence, the project cost consideration is justified. Assessment team checked the actual project cost and still the project do not breach the benchmark. The sensitivity analysis below confirms the same. Since the actual cost is considered, there is no way the cost can go up and thus the same is assessed to be correct.

The profitability of the project, which forms the basis for IRR calculation is based on installed capacity, PLF, tariff rate, O&M cost, depreciation, and taxation.

C) Assessment of Plant Load Factor (PLF):

PP considered the Plant load factor as per DPR: -

Site Name	Project Proponent	Project Capacity (MW)	PLF (%)= 3rd party report	DPR Date
Spectra Solar Park Ltd _35	Spectra Solar Park Limited	35 MW	20.08	March 2018

MW AC Grid-Tied				
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D) Assessment of Electricity Tariff:

Tariff rate as per DPR and PPA: -

Site Name	Project Owner	Tariff Rate (as per DPR)	Tariff Rate (as per PPA)	PPA Date
Spectra Solar Park Ltd _35 MW AC Grid-Tied	Spectra Solar Park Limited	0.139 USD/KWh	0.139 USD/KWh	04-November- 2018

Verification team assessed the tariff and found that same value was available during decision making and in conformity with guidance Appendix of Methodological tool “Investment Analysis” Version 12.0. Furthermore, assessment team has also checked the actual tariff as mentioned in sample electricity invoices. IRR is still below benchmark with the consideration of PPA signed which is valid for total operational lifetime of the project.

E) Assessment of O& M cost:

O&M as per Detailed Project Report: -

Site Name	Project Owner	Project Capacity (MW _{AC})	O&M Cost (In Million) – without tax	DPR Date
Spectra Solar Park Ltd _35 MW AC Grid-Tied	Spectra Solar Park Limited	35	0.297	March 2018

According to Appendix of Methodological tool “Investment Analysis” Version 12, the cost should be based on the input parameters available at the time of decision making and the PP has submitted O & M agreement (incorporating cost of initial year) supporting this consideration. Therefore, considering the above assessment, verification team concluded that the O&M cost considered from respective agreement in the computation of financial indicator is in conformity with guidance Appendix of Methodological tool “Investment Analysis” Version 12.

F) Cross checking parameters:

Name of the parameter	VVB assessment								
Project Cost	The details for the proposed project activity are given below: <table><tr><th>Project Owner</th><th>Project Capacity (MW)</th><th>Project Cost (In Million)</th><th>Project cost (in Million per MW)</th></tr><tr><td>Spectra Solar Park Limited</td><td>35</td><td>52.064 Millions</td><td>1.48 Million/MW</td></tr></table> The project cost has been considered from DPR and was available at the time decision made for the project activity. The assessment team has also checked the actual cost of each project site from the CA certificate and found that, the reduction in project cost is within 10% range of sensitivity analysis. Since the comparison is done with actual project cost and hence, increase of the same in future is not possible. Thus, the project activity is additional with actual project cost. The assessment team also checked the state tariff orders and found that project cost considered for the project is found to be appropriate. Based on the title report "Indicative Tariff for Utility-Scale Solar IPP in Bangladesh by Center for Renewable Energy Services Ltd. (CRESL)" the indicative cost is 1.34 million USD/MW(AC) as against the unit project cost of 1.48 million USD/MW(AC) for the current project activity. The cost difference of 10% owes to higher land development cost as the project is developed on rived bed. Recent PDD registered from Bangladesh has used 1.9 million USD/MW(AC) for estimation of return- https://registry.verra.org/app/projectDetail/VCS/3545 Based on sectoral scope expert and local knowledge, the project cost considered as per DPR for the proposed project activity is found to be appropriate for solar projects. Also, since the actual cost is available to assessment team and IRR is still within benchmark and thus the same is acceptable. The IRR as per the assumption from the DPR is as follows:	Project Owner	Project Capacity (MW)	Project Cost (In Million)	Project cost (in Million per MW)	Spectra Solar Park Limited	35	52.064 Millions	1.48 Million/MW
Project Owner	Project Capacity (MW)	Project Cost (In Million)	Project cost (in Million per MW)						
Spectra Solar Park Limited	35	52.064 Millions	1.48 Million/MW						

	<table><tr><th>Site Name</th><th>Project Owner</th><th>Project Capacity (MW)</th><th>Project Cost (In Million)</th><th>IRR</th><th>Benchmark</th></tr><tr><td>Spectra Solar Park Ltd _35 MW AC Grid-Tied</td><td>Spectra Solar Park Limited</td><td>35 MW</td><td>52.064</td><td>10.84 %</td><td>18.21%</td></tr></table>	Site Name	Project Owner	Project Capacity (MW)	Project Cost (In Million)	IRR	Benchmark	Spectra Solar Park Ltd _35 MW AC Grid-Tied	Spectra Solar Park Limited	35 MW	52.064	10.84 %	18.21%
	Site Name	Project Owner	Project Capacity (MW)	Project Cost (In Million)	IRR	Benchmark							
	Spectra Solar Park Ltd _35 MW AC Grid-Tied	Spectra Solar Park Limited	35 MW	52.064	10.84 %	18.21%							
	The IRR as per the actual project cost is defined as below:												
<table><tr><th>Site Name</th><th>Project Owner</th><th>Project Capacity (MW)</th><th>Actual project cost as per the CA Certificate</th><th>IRR</th><th>Benchmark</th></tr><tr><td>Spectra Solar Park Ltd _35 MW AC Grid-Tied</td><td>Spectra Solar Park Limited</td><td>35 MW</td><td>48.444</td><td>13.35 %</td><td>18.21%</td></tr></table>	Site Name	Project Owner	Project Capacity (MW)	Actual project cost as per the CA Certificate	IRR	Benchmark	Spectra Solar Park Ltd _35 MW AC Grid-Tied	Spectra Solar Park Limited	35 MW	48.444	13.35 %	18.21%	
Site Name	Project Owner	Project Capacity (MW)	Actual project cost as per the CA Certificate	IRR	Benchmark								
Spectra Solar Park Ltd _35 MW AC Grid-Tied	Spectra Solar Park Limited	35 MW	48.444	13.35 %	18.21%								
As described above actual project cost with benchmark, the project is still additional. Since the comparison is done with actual project cost, the increase of the same in future is not possible. Thus, assessment team is of the opinion that project is still additional with the consideration of actual project cost for the project activity.													
O&M cost and Escalation in the operational expense = 5.72%	The details for the proposed project activity are given below: <table><tr><th>Project Owner</th><th>Project Capacity (MW)</th><th>O&M Cost (In Million/per MW)</th></tr><tr><td>Spectra Solar Park Limited</td><td>35 MW</td><td>0.2975</td></tr></table>						Project Owner	Project Capacity (MW)	O&M Cost (In Million/per MW)	Spectra Solar Park Limited	35 MW	0.2975	
Project Owner	Project Capacity (MW)	O&M Cost (In Million/per MW)											
Spectra Solar Park Limited	35 MW	0.2975											
The O&M cost has been considered from DPR and was available at the time decision made for the project activity.													

The assessment team has also checked the actual O&M contract for the project activity and found the changes in O&M cost is within threshold limit. Thus, the project activity is additional with actual O&M cost.

Site Name	Project Owner	Project Capacity (MW)	O&M Cost (In Million/per MW) (Without tax) - Actual
Spectra Solar Park Ltd _35 MW AC Grid-Tied	Spectra Solar Park Limited	35 MW	0.3494

The assessment team also checked the Bangladesh tariff orders and found that O&M cost and its escalation considered for project is found to be appropriate.

IRR value as per the assumptions from the DPR is as below:

Project Owner	Project Capacity (MW)	O&M Cost (In Million/per MW)	IRR	Benchmark
Spectra Solar Park Limited	35 MW	0.2975	10.84%	18.21%

IRR value based on the actual O&M agreements signed is as below:

Site Name	Project Owner	Project Capacity (MW)	O&M Cost (In Million) (Without tax)- Actual	IRR	Benchmark
Spectra Solar Park Ltd _35 MW AC Grid-Tied	Spectra Solar Park Limited	35 MW	0.3494	10.44 %	18.21%

	Benchmark for the project as described above along with actual O&M value, the project is still additional. Based on sectoral scope expert and local knowledge, the project O&M cost and its escalation considered as per O & M agreement for the proposed project activity is found to be appropriate for solar projects. Also, the O&M cost is available to assessment team and IRR is still within benchmark, thus, the same is acceptable.																				
Tariff	The details for the proposed project activity are given below:- <table><tr><th>Site Name</th><th>Project Owner</th><th>Project Capacity (MW)</th><th>Tariff Rate (as per DPR)</th><th>Tariff Rate (as per PPA)</th></tr><tr><td>Spectra Solar Park Ltd _35 MW AC Grid-Tied</td><td>Spectra Solar Park Limited</td><td>35 MW</td><td>0.139 USD/KW H</td><td>0.139 USD/KW H</td></tr></table> The Tariff rate has been considered from DPR and the same was available at the time decision made for the project activity. The assessment team has also checked the actual PPA for each project site and found there are no changes in tariff rate and is within threshold limit. Thus, the project activity is additional with actual Tariff rate. The tariff considered is levelized tariff and hence there is no any escalation. This is found to be appropriate and accepted. IRR value as per the assumptions from the DPR is as below: <table><tr><th>Site Name</th><th>Project Owner</th><th>Tariff Rate (as per PPA)</th><th>IRR</th><th>Benchmark</th></tr><tr><td>Spectra Solar Park Ltd _35 MW AC Grid-Tied</td><td>Spectra Solar Park Limited</td><td>0.139 USD/KWH</td><td>10.84 %</td><td>18.21%</td></tr></table>	Site Name	Project Owner	Project Capacity (MW)	Tariff Rate (as per DPR)	Tariff Rate (as per PPA)	Spectra Solar Park Ltd _35 MW AC Grid-Tied	Spectra Solar Park Limited	35 MW	0.139 USD/KW H	0.139 USD/KW H	Site Name	Project Owner	Tariff Rate (as per PPA)	IRR	Benchmark	Spectra Solar Park Ltd _35 MW AC Grid-Tied	Spectra Solar Park Limited	0.139 USD/KWH	10.84 %	18.21%
Site Name	Project Owner	Project Capacity (MW)	Tariff Rate (as per DPR)	Tariff Rate (as per PPA)																	
Spectra Solar Park Ltd _35 MW AC Grid-Tied	Spectra Solar Park Limited	35 MW	0.139 USD/KW H	0.139 USD/KW H																	
Site Name	Project Owner	Tariff Rate (as per PPA)	IRR	Benchmark																	
Spectra Solar Park Ltd _35 MW AC Grid-Tied	Spectra Solar Park Limited	0.139 USD/KWH	10.84 %	18.21%																	

	during decision making context and thus acceptable to the assessment team.
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Sensitivity analysis:

The Guidance on Investment analysis/09/ requires the robustness of the conclusion arrived at to be proved through a sensitivity analysis by varying the critical assumptions to a reasonable variation. The project developer has identified Plant Load Factor (PLF), Project cost, tariff rate and O&M cost as critical assumptions. These critical parameters constitute more than 20% of either total project costs or total project revenues. The sensitivity analysis reveals that even under more favorable conditions, the IRR would not cross the benchmark return as given in the following table:

Sensitivity Analysis	Equity IRR			
Variation %	-10%	Normal	10%	Breaching Value
PLF	7.17%	10.84%	14.56%	19.75%
O&M	11.06%	10.84%	10.61%	-354.14%
Project Cost	14.74%	10.84%	8.21%	-15.84%
Tariff Rate	7.17%	10.84%	14.56%	19.75%

The results of sensitivity analysis show that even with a variation of +10% & -10% in project cost, O&M cost, PLF and Tariff Rate, Equity IRR is significantly lower than the benchmark. And it is evident from the results given above; the project remains additional even under the most favorable conditions.

Assessment team also confirmed the breaching values for individual parameters (=Individual project owners) and thus confirms that the project is still additional

Site Name	Project Owner	Project Capacity (MW)	PLF in DPR	Tariff order PLF	Variation in PLF	Breaching Value for PLF
Spectra Solar Park Ltd _35 MW AC Grid-Tied	Spectra Solar Park Limited	35 MW	18%	19%	5.26%	26.93%

Site Name	Project Owner	Project Capacity (MW)	DPR Project Cost	Actual Cost from CA Certificate	Variation in project cost	Breaching value for Project Cost
Spectra Solar Park Ltd _35 MW AC Grid-Tied	Spectra Solar Park Limited	35 MW	52.064	48.444	-6.95%	-15.84%

Site Name	Project Owner	Project Capacity (MW)	DPR Tariff USD/ KWh	PPA Tariff USD/ KWh	Variation in Tariff	Breaching value in Tariff Rate
Spectra Solar Park Ltd _35 MW AC Grid-Tied	Spectra Solar Park Limited	35 MW	0.139	0.139	0%	19.75%

Site Name	Project Owner	Project Capacity (MW)	DPR O&M cost	Actual O&M cost	Variation in O&M	Breaching value in O&M
Spectra Solar Park Ltd _35 MW AC Grid-Tied	Spectra Solar Park Limited	35 MW	0.2975	0.3494	-14.85%	-354.14%

The above discussions show that solar power development is not a financially attractive; hence the project activity is additional and the assessment team considers the approach and calculations acceptable as per the requirements in the methodological tool.

Common practice Analysis

In accordance with paragraph 57 of the “Tool for the demonstration and assessment of additionality” – (Version 07.0.0), **Error! No se encuentra el origen de la referencia.** the project has been subjected to an analysis of the extent to which the proposed project type (e.g.

technology or practice) has already diffused in the relevant sector and region.” The project meets the criteria of tool “Common Practice”, Version 3.1/9/ as follows:

- *Applicable geographical area:* As confirmed from the commissioning certificate, the state of **Bangladesh** is considered as applicable geographical area.

As per the requirement of paragraph 9 of the tool “Common Practice”, Version 3.1/09/ “the applicable geographical area should be the entire host country. If the project participants opt to limit the applicable geographical area to a specific geographical area (such as province, region, etc.) within the host country, then they shall provide justification on the essential distinction between the identified specific geographical area and rest of the host country.”

- *Measure:* Project has been confirmed to be renewable energy generation activity, which has also been confirmed through the commissioning certificate and power purchase agreement.
- *Technology & Output:* As confirmed from the commissioning certificate, the project Owner is generating power using solar panels. The power generated by solar panels is fed to the Bangladesh Electricity Grid.

As According to the methodological tool “Common Practice”, Version 3.1|**Error! No se encuentra el origen de la referencia.**, the step wise demonstration of common practice analysis is verified as follows :

Step 1 :- Calculate applicable capacity or output range as +/-50% of the design capacity or output of the proposed project activity :

The size of the project activity is 35 MWac, so the applicable output range as defined in the step is +/-50% of the capacity of the proposed project activity. This output range will come in a capacity range of 17.5 MW to 52.5 MW

Step 2 :- Identify similar projects (both CDM and non-CDM) which fulfil all of the following conditions:

Identification of the similar projects (CDM and non-CDM) is carried out as per sub-steps of Step (2) as follows:

- As the project is located Bangladesh therefore, projects in the geographical area of Bangladesh have been chosen for analysis.
- The project activity is a green-field solar power project and uses measure (b) “Switch of technology with or without change of energy source including energy efficiency improvement as well as use of renewable energies”. Therefore, projects applying same measure (b) are candidates for similar projects.
- The energy source used by the project is solar. Hence, only solar energy projects have been considered for analysis.
- The project produces electricity; therefore, all power plants that produce electricity are candidates for similar projects.
- The capacity range of the projects is within the applicable capacity range from 17.5 MW to 52.5 MW.
- As per the methodological tool the projects that have started commercial operation before the project design document (CDM-PDD) is published for global stakeholder consultation or before the start date of proposed project activity, are to be selected for consideration. The

start date however resembles to start date definition of CDM project activity which is “the date on which the project participants commit to making expenditures for the construction or modification of the main equipment or facility (e.g. a wind turbine), or for the provision or modification of a service (e.g. distribution of energy-efficient light bulbs, change of transport management system), for the CDM project activity or CPA. Where a contract is signed for such expenditures.

Findings of analysis of Step 2

Numbers of Similar projects identified, which fulfil above-mentioned conditions have been identified from the annual report of Bangladesh Power Development Board. Based on 'the published information there is only two solar power plant within the capacity range.

1. Teknaf 20MW PP (Solartech)/05/
2. HDFC Solar 50 MW, Mymensing/05/

Step 3: within the projects identified in Step 2, identify those that are neither registered CDM project activities, project activities submitted for registration, nor project activities undergoing validation. Note their number N_{all}

Project activities, which have got registered or are under project verification with CDM/VCS/GS/GCC have been excluded in this step. After excluding the registered and under project validation projects the total number of projects.

$N_{all} = 0$

Step 4 : within similar projects identified in Step 3, identify those that apply technologies that are different to the technology applied in the proposed project activity. Note their number N_{diff} .

The Project is deployed under preferential tariff regime (Investment climate on the date of the investment decision, inter alia- Promotional policies) so solar power project that are not covered under preferential tariff/promotional policy are considered as different from the project activity. However, there is only 1 project so project with different measures is considered as zero.

$N_{diff} = 0$

Step 5 : Calculate factor $F = 1 - N_{diff}/N_{all}$ representing the share of similar projects (penetration rate of the measure/technology) using a measure/technology similar to the measure/technology used in the proposed project activity that deliver the same output or capacity as the proposed project activity.

$$F = 1 - N_{diff} / N_{all}$$

$$F = 1$$

So, F is greater than 0.2

$$N_{all} - N_{diff} = 0$$

So $N_{all} - N_{diff}$ is less than 3

As per the Tool for common practice analysis, version 03.1, the proposed project activity is a common practice within a sector in the applicable geographical area if both the following conditions are fulfilled:

- (a) the factor F is greater than 0.2, and
- (b) $N_{all} - N_{diff}$ is less than 3.

Since the value of factor $N_{all} - N_{diff}$ for the proposed project activity is 1 which is less than 3, the project activity is not a “common practice” within sector in the applicable geographical area

3.4.6 Quantification of GHG Emission Reductions and Removals

Assessment team checked the baseline, project and leakage calculation and confirm that the evaluation of baseline, project and leakage is as per the approved methodology and formula used to calculate the same is correct. The detail analysis is as below:

Baseline Emissions: As per para 47 of ACM0002, version 21.: ACM0002 Grid-connected electricity generation from renewable sources ---Version 21.0/09/, the Baseline emissions for other systems are the product of amount electricity displaced with the electricity produced by the renewable generating unit and an emission factor.

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$$

where:

BE_y	Baseline emissions in year y (t CO ₂ /yr)
$EG_{PJ,y}$	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)
$EF_{grid,CM,y}$	Combined margin CO ₂ emission factor for grid connected power generation in year y calculated using TOOL07 (t CO ₂ /MWh)

Since the project is a green-field power plant therefore calculation of quantity of net electricity generation ($EG_{PJ,y}$) is carried out in accordance to section 5.5.1.1, paragraph 49 of the approved consolidated Methodology ACM0002, V 21.0, :

$$EG_{PJ,y} = EG_{facility,y} \text{-----Equation (12)}$$

$EG_{PJ,y}$	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)
$EG_{facility,y}$	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)

Project Emissions: -

The project emission calculation as per para 35 of ACM0002 version 21

For most renewable energy power generation project activities, $PE_y = 0$. However, some project activities may involve project emissions that can significant. These emissions shall be accounted for as project emissions by using the following equation:

$$PE_y = PE_{FF} + PE_{GP} + PE_{HP,y} + PE_{BESS,y} \text{ -----Equation (1)}$$

Where:

PE_y	Project emissions in year y (t CO _{2e} /yr)
PE_{FF}	Project emissions from fossil fuel consumption in year y (t CO _{2e} /yr)
PE_{GP}	Project emissions from the operation of geothermal power plants due to the release of non-condensable gases in year y (t CO _{2e} /yr)
$PE_{HP,y}$	Project emissions from water reservoirs of hydro power plants in year y (t CO _{2e} /yr)
$PE_{BESS,y}$	Project emissions from charging of a BESS using electricity from the grid or from fossil fuel electricity generators (t CO _{2e} /yr)

The project activity is a solar power project without storage system, and in accordance to ACM0002 version 21 /09/, and does not include any fossil fuel consumption, therefore in accordance to Para 35,

$$PE_y = 0$$

Leakage Emissions: - As per the methodology ACM0002 v21, para 61, no other leakage emissions are considered. In accordance to the methodology "The emissions potentially arising due to activities such as power plant construction and upstream emissions from fossil fuel use (e.g. extraction, processing, transport etc.) are neglected".

Therefore, the net emission reduction is calculated as

$$ER_y = BE_y - PE_y$$

Year	Estimated baseline emissions or removals (tCO _{2e})	Estimated project emissions or removals (tCO _{2e})	Estimated leakage emissions (tCO _{2e})	Estimated net GHG emission reductions or removals (tCO _{2e})
Year 1	37,827	0	0	37,827
Year 2	37,562	0	0	37,562

Year 3	37,299	0	0	37,299
Year 4	37,038	0	0	37,038
Year 5	36,779	0	0	36,779
Year 6	36,521	0	0	36,521
Year 7	36,266	0	0	36,266
Total	259,292	0	0	259,292

3.4.7 Methodology Deviations

Assessment team confirms that no methodology deviation is applicable for the present project activity.

3.4.8 Monitoring Plan

Assessment team checked the monitoring practice onsite and checked the guideline of respective State electricity regulatory commission. The detail analysis is as below:

Parameters determined ex-ante:

Baseline emission factor of Bangladesh Grid is established ex-ante based on Tool to calculate the grid emission factor, using a combined approach consisting 75 % operating margin and 25 % build margin. The emission coefficient from official data published in Clean Development Mechanism (CDM) Baseline, Power and Energy Sector, published by Department of Environment, Ministry of Environment and Forest, Bangladesh and Annual Report published by “Bangladesh Power Development Board. Database available to the project participant at the time of submission of joint VCS PD & MR for joint validation & Verification and global stakeholder’s consultation process. The assumption was verified by the validation team and found to be correct.

Parameters determined ex-post:

The parameters monitored ex-post involves quantity of net electricity displaced as a result of the implementation of the project activity.

As per the Joint VCS PD & MR Version 04, Meter Reading Report will be the source of data during verification. The VVB will use the same source for verification of emission reductions.

In accordance with the methodology requirement, net electricity supplied by the project activity is obtained from Meter Reading Report issued by Bangladesh Power Development Board (BPDB) which provide input values used for calculation of $EG_{Facility,y}$ by the project activity and forms the basis for emission reduction calculation. Electricity export to the grid and import from the grid is metered by main energy meter. The main meter reading is taken jointly on a fixed day of every

month for the preceding month at the delivery point and signed by the representatives of project developer and customer.

The agency is experienced in the monitoring system and is managing O&M /14/of numerous other solar projects. The validation team therefore is of the opinion that the project participant through the developer/O&M agency can implement the monitoring plan in the context of the project activity.

It is in line with the BERC notification, 2019/09/ (after PPA signing) the meter will be tested for accuracy at the interval of 1 year. In line with the Gazette Notification by Bangladesh Energy Regulatory Commission (BERC) dated 31 December, 2019¹⁰, the calibration of metering system is required to be carried out at intervals after 5 years (Section 13.6.10), along with periodical yearly testing of meter and associated equipment's.

However, with reference to PPA signed/14/ in November 2018, project proponent needs to undertake inspection, testing, and calibration of the Metering System upon installation under observation of BPDB (Bangladesh Power Development Board). Thereafter, BPDB shall test the Metering System at intervals of not less than 180 (one hundred eighty) Days.

It is reported that the data will be kept for 2 years following the end of the crediting period or till the last issuance of VCU's for the project activity whichever occurs later.

The responsibilities and authorities of project management, data handling and recording, measurement methods and QA/QC procedure have been systematically established and formalized and the same was verified during the interview.

3.5 Non-Permanence Risk Analysis

Not applicable for the present project activity.

4 VERIFICATION FINDINGS

4.1 Project Implementation Status

During the remote assessment, it was concluded that the project is implemented as per the requirement of the VCS joint PD & MR and mentioned monitoring plan, during the current monitoring period; it was observed that no unforeseen incident/event evolved which can impact

¹⁰

http://berc.portal.gov.bd/sites/default/files/files/berc.portal.gov.bd/page/a250b6fc_8bcf_4c96_bb20_3c3de230467a/2020-03-01-17-04-6c286949cb014aac31754ffa3e86bac2.pdf

the operation of the project activity which was verified from breakdown records. The project undergone continuous operation and only scheduled maintenance is observed as per the manufactures specification which is acceptable to the assessment team and evident from Meter Reading Reports.

The project activity is an installation of Solar Photovoltaic Panels in Manikganj district of Dhaka region of Bangladesh. The sole purpose of project activity is to install solar power plant for electricity selling purpose, same is confirmed by the verification team during the remote assessment.

Verification team confirmed that the location of the project activity including the coordinates is same as mentioned in the VCS joint PD & MR.

Assessment team checked the commissioning certificate/01/ and confirmed that the dates of Commission for the solar power plant are correct. Assessment team also conform during remote interview with the PPs representatives that there is no change in project design and the project is implemented as per the description provided in the VCS PD & MR. The project boundary includes the electricity generation equipment at the project site. Assessment team also checked the technical details of the solar power plant installed from submitted documents and site photographs/reference/ submitted by PP. The assessment team confirmed that there is no proposed or actual change to the project design during this monitoring period. The project design as mentioned in the VCS joint PD & MR is implemented and thus the same is acceptable to the assessment team. All required monitoring equipment's and procedures as mentioned in the VCS joint PD & MR version 04 are available and implemented in an appropriate manner.

The project is not participating in any other environmental credits or other GHG programs and has not been rejected by any GHG programs. The project activity is not participating in any other Emission trading programme and other binding limits. The declaration/17/ mentioning the same has been submitted by the PP to the VVB. The project has implemented activities that contribute to sustainable development as described in the joint PD & MR and the same has been verified during the remote assessment and interviews.

4.2 Accuracy of GHG Emission Reduction and Removal Calculations

Means of verification	The verification team assessed whether the data and calculations of GHG emission reductions achieved resulting from the Joint VCS PD & MR. The verification team has checked whether calculations of baseline GHG emissions, project GHG emissions and leakage GHG emissions have been carried out in accordance with the formulae and methods described in the monitoring plan of the Joint VCS PD & MR.
Findings	CAR 09 raised on this section and closed successfully. Please refer Appendix 1 for further details

Conclusion	The baseline Emissions for a given year is calculated by multiplying the energy baseline (EB) with the regional grid emission factor. The grid in this case would be the 'Bangladesh Grid' Formula Used: - $BE_y = EG_{Facility,y} \times EF_{grid,CM,y}$ $BE_y = 54,066.8544 \text{ MWh} \times 0.6145 \text{ tCO}_2/\text{MWh}$ $= 33,223 \text{ tCO}_2 \text{ (round down) (for this monitoring period)}$ EG_{Facility,y} (Quantity of net electricity generation supplied by the project plant/unit to the grid in year y - MWh/year): The verification team has checked the entire monthly meter reading report and invoices applicable for the monitoring period as per the project activity applied for verification and found all the electricity generation units are monitored and recorded as per the monitoring plan mentioned in the Joint VCS PD & MR. The Net electricity supplied to the grid by the project activity is specified in the billing statement. The net generation measured by Main Meter is considered for the purpose of billing settlement. During monitoring period, value is taken as 54,066.8544 MWh. The verification team has crosschecked the emission reduction sheet and monitoring report data with the Meter reading report and invoice bills and found all the values are matching. Bidirectional Energy meters were used at each instances location to measure energy net generated and supplied to Bangladesh Grid. During current monitoring period, In line with the guidelines of BERC, the meter has been tested/10/ and the accuracy level is assessed to be within the specified limit. Test report of energy meter dated 09/02/2021 and Inspection & test report of energy meter dated 19/03/2022 is verified by validation and verification team. Thus accepted.
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4.3 Quality of Evidence to Determine GHG Emission Reductions and Removals

Means of verification	The verification team checked the break down log for the monitoring period. During the verification site visit each meter location of the solar plants is also checked. The Calibration details of the monitoring meters are also checked with calibration certificates.
Findings	CAR 09 was raised on this section and closed successfully. Please refer Appendix 1 for further details

Conclusion	The verification team checked the break down log for the monitoring period. During the remote audit, meter location of project is also checked. Quantity of net electricity displaced as a result of the implementation of this project activity is the monitoring parameter as per the Joint VCS PD & MR/01/. The metering arrangement is bi-directional energy meters (Main meter) of accuracy class 0.2s at each project site. These electricity meters are being used by Spectra Solar Park Limited for Meter Reading Report preparation. The details like make, Serial number, Calibration dates etc. are provided in appendix 05 of this report. Verification team confirms that all the energy meters (main meter) installed at project site are of accuracy class of 0.2s and are calibrated as per the calibration frequency mentioned in monitoring plan in VCS PD & MR i.e. The calibration frequency of meters is as per the BERC regulations/09/. No delayed calibrations were observed in the project activity for this monitoring period. All the meters are of same accuracy class i.e., 0.2s as per the requirement of the VCS approved PD & MR. In future meter change may happen with same accuracy class or with better accuracy class. Interview during remote visit with O&M personnel also confirms the same. The break down log is checked and there is no major breakdown during the monitoring period. No unforced error observed. No sampling procedure applied for monitoring of the data parameter and entire documents were checked by the assessment team to arrive at positive verification conclusions. The monitoring plan is followed at the project site. The monitoring meters were calibrated in line with the registered monitoring plan and there was no delay in calibration observed. Thus, assessment team concluded that the evidences are sufficient in quantity, and appropriate for the quality, to determine the GHG reductions and removals. Assessment team checked the calculation of Estimated VCUs (33,084 tCO₂e) vs. Achieved VCUs (33,223 tCO₂e). As per the VCS joint PD & MR the amount of VCUs estimated annually is 37,042 tCO₂e. Estimated value of VCUs has been calculated are as follows: Total no of days under the current monitoring period = 326 Annual estimated value as per the PDD = 37,042 tCO₂ Estimated VCUs with reference to PDD for current monitoring period = (37,042 tCO₂/365 days)*326 days = 33, 084 tCO₂ The actual emission reduction has been observed to be 0.42% increase (Overall project) in comparison with the ex-ante estimations due to high plant load factor (PLF) and other parameters like climatic conditions like solar
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	irradiance, temperature etc which are not in the control of project proponent. Same was confirmed with PP during remote assessment.
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5 VALIDATION AND VERIFICATION CONCLUSION

LGA Technological Center, S.A. (also referred to as Applus+ Certification), contracted by Spectra Solar Park Limited to perform a joint validation and verification of the VCS project activity “Spectra Solar Park Ltd _35 MW AC Grid-Tied by Spectra Solar Park Limited”.

The joint validation and verification process were performed on the basis of all guidance and criteria as provided in VCS Standard (version 4.4), VCS Program Guide (version 4.3) and Registration & Issuance Process (version 4.3).

The management of the project participant/owner is responsible for the preparation of the GHG emissions data and the reported/estimated GHG emissions reductions on the basis set out within the project’s Monitoring Plan in the Joint VCS PD & MR and the approved methodologies; “ACM0002 Grid-connected electricity generation from renewable sources --- Version 21.0/09/.

The conclusions of validation and verification process can be individually summarized as follows:

Validation Conclusion:

Our Validation approach was based on the requirements as defined under the Kyoto Protocol, Marrakesh accord, as well as those defined by the CDM Executive Board and VCS board. Our approach is risk-based, drawing on an understanding of the risks associated with estimated GHG emissions data and the controls in place to mitigate these. The validation can confirm that:

- The projects description compliance with, the requirements of Article 12 of the Kyoto Protocol/06/, the CDM Modalities and Procedures as agreed in the Marrakech Accords under decision 3/CMP.1, the annexes to this decision, subsequent decisions and guidance made by COP/MOP & CDM Executive Board and other relevant rules, including the Host Country legislation and sustainability criteria along with VCS Program Guide,v4.3 and standard version 4.4/09/.
- The project’s baseline and additionality are assessed against ““ACM0002 Grid-connected electricity generation from renewable sources ---Version 21.0. and Tool for the demonstration and assessment of additionality – Version 07.0.0./09/

- The project's monitoring plan is assessed against "ACM0002 Grid-connected electricity generation from renewable sources --Version 21.0. /09/
- A risk-based approach has been followed to perform this validation activity. The review of the project description and additional documents related to baseline and monitoring methodology; the subsequent background investigation, follow-up interviews with Project Owner have provided LGAI Technological Center S.A. (Applus+ Certification) with sufficient evidence for positive validation opinion as per the requirement of VCS.

The project is expected to generate 259,292 tCO₂e during the length of entire crediting period (12-March-2021 to 11-March-2028).

Verification Conclusion: -

Our Verification approach was based on the requirements as defined under the Kyoto Protocol, Marrakesh accord, as well as those defined by the CDM Executive Board. Our approach is risk-based, drawing on an understanding of the risks associated with reporting GHG emissions data and the controls in place to mitigate these. The verification can confirm that

- The project is operated as planned and described in the project document;
- The monitoring plan is as per the applied methodology;
- The monitoring process in Monitoring Report is as per the joint PD & MR.
- The development and maintenance of records and reporting procedures are in accordance with the monitoring plan;
- The installed equipment being essential for generating emission reduction runs reliably and is calibrated appropriately
- The monitoring system is in place and generates GHG emission reductions data;
- The GHG emission reductions are calculated without material misstatements.
- No limitation observed for the present verification

Verification period: 12-March-2021 to 31-January-2022 (first and last date included)

Verified GHG emission reductions and removals in the above verification period:

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
12-03-2021 to 31-12-2021	30,336	0	0	30,336

1-1-2022 to 31-1-2022	2,887	0	0	2,887
Total	33,223	-	-	33,223

APPENDIX 1: DOCUMENTS REVIEW UNDER VALIDATION & VERIFICATION

No.	Author	Title	References to the document	Provider
1.	Bangladesh Power Development Board	Commissioning certificate	Ref. No. 27.11.0000.101.14.013.21-793	Project participant
2.	Applus	Contract of the PP with the DOE	Contract of the PP with the DOE	Project participant
3.	NA	Technical specifications	Technical specifications of each SPV from manufacturers	Project participant
4.	PP	Draft Joint VCS PD&MR Version 01 Final Joint VCS PD & MR version 04	20-April-2022 24-February-2023	Project participant
5.	PP	Emission Calculation sheet- version 01.1- Estimated & Actual ER Emission Calculation sheet- version 02.0- Estimated and actual ER IRR spreadsheet, V3.0 Common Practice Analysis spreadsheet	20-April-2022 22-January-2023 - -	Project participant
6.	PP	- CDM validation and verification standard for project activities, Version 03.0 - Kyoto Protocol (1997)	https://cdm.unfccc.int/Reference/Standards/index.html https://unfccc.int/process-and-meetings/the-kyoto-protocol/what-is-the-kyoto-protocol/kyoto-protocol-	Project participant

No.	Author	Title	References to the document	Provider
			targets-for-the-first-commitment-period	
7.	NA	The operational lifetime of the project activity from the manufacturer= (Technical specifications)	Manufacturer technical specifications	Project participant
8.	NA	Reference link is provided.	UNFCCC www.cdm.unfccc.int	Independent Search
9.	UNFCCC VERRA	Tools/ guidelines used in the project activity	UNFCCC CDM web site • UNFCCC Methodology: ACM0002 Grid-connected electricity generation from renewable sources --Version 21.0. (https://cdm.unfccc.int/UserManagement/FileStorage/ZPFJL010U2RYC6N3HASIXV7K84QBG9) • Tool 01 - Tool for the demonstration and assessment of additionality - Version 07.0.0 (https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-01-v7.0.0.pdf) • Tool 07 - Tool to calculate the emission factor for an electricity system -Version 07.0 (https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-07-v7.0.pdf) • Tool 24 - Common practice - Version 03.1 (https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-24-v1.pdf) • Glossary of CDM terms version 11 (https://cdm.unfccc.int/Reference/Guidclarif/glos_CDM.pdf) • VCS joint validation and verification report template Version 4.1. (https://verra.org/programs/verified-carbon-standard/vcs-program-details/) • VCS standard V4.4 (updated on 17-January-2023) • VCS Program Guide V4.3 (updated on 17-January-2023) (https://verra.org/programs/verified-carbon-standard/vcs-program-details/#rules-and-requirements) Environmental Approval with ref no 21-	UNFCCC

No.	Author	Title	References to the document	Provider
			67955 BERC Notification http://berc.portal.gov.bd/sites/default/files/files/berc.portal.gov.bd/page/a250b6fc_8bcf_4c96_bb20_3c3de230467a/2020-03-01-17-04-6c286949cb014aac31754ffa3e86bac2.pdf)	
10.	Bangladesh Power Development Board (BPDB)	Calibration details of the project activity undergoing verifications	Please refer to Appendix 5	Project participant
11.	PP	Plant records + log book	Meter Reading Report (JMR + Invoices) covering entire monitoring period i.e. from 11-March – 2021 to 31-January-2023	Project participant
12.	PP	VCS Declaration	Declaration from PP for Participation under Other GHG Programs	Project participant
13.	PP	LSC	Local stakeholders' consultation supporting; Notices, MoM's & photographs Feedback register etc.	Project participant
14.	PP	- PPA - O & M Agreement	Power purchase agreement signed between Spectra Solar Park Limited and Bangladesh Power Development Board (BPDB) dated on 04-November-2018 Project participant	Project participant
15.	PP	Site photo photographs along with installed major equipment's/name plates	-	Project participant
16.	PP	Local Stakeholder consultation	- Snapshots and supportive documents, Feedback Forms etc - ESIA consultant Report	Project participant
17.	PP	Letter of Undertaking	For no double counting and part into any emission trading program and not received credits under other registry	Project participant
18.	NA	Risk Analysis for conducting Remote Audit	Dated: 12-Decemeber-2022	VVB

APPENDIX 2: CLARIFICATION REQUESTS, CORRECTIVE ACTION REQUESTS, FORWARD ACTION REQUESTS (CAR/CL/FAR)

Remaining FAR from validation and/or previous verifications

FAR ID	xx	Section no.		Date: DD/MM/YYYY
Description of FAR				
N/A				
Project participant response				Date: DD/MM/YYYY
Documentation provided by project participant				
DOE assessment				Date: DD/MM/YYYY

Table 1. CL from this verification

CL ID	01	Section no.	1.4	Date: 05/01/2023
Description of CL				
Project Owner to clarify and justify with the evidence:				
Under Section 1.1 of submitted Joint PD Monitoring Report, the applied value of annual average generation i.e.60,280 MWh/year, mentioned as 7 year average while under parameters monitoring part it is taken as 3 year average.				
Project participant response				Date: 18/01/2023
- The average generation outlined under section 5.2 of the joint PD-MR was for 7 years and 3 -years referred in the web-hosted document was a typographic error. - The value applied in section 5.2 is revised to expected first year generation i.e. 61,558 MWh/year instead of average annual generation for 7 years.				
Documentation provided by project participant				
Revised joint PD-MR				

DOE assessment	Date: 31/01/2023
Under revised Joint PD Monitoring Report, the typographic error has now been removed instead of 7 years instead of 3 years. However, the generation should be average annual of all the seven years shall be incorporated under the section 5.2. Therefore, this observation is still open. Also, request to incorporate sample calculation of estimated emission reductions by taking annual average generation under section 4.1. Hence, finding (CL – 01) is currently open	
Project participant response	Date: 15/02/2023
1. Section 5.2 have been revised to include the average 7-year generation value 2. Sample calculation (estimation of baseline emission) has been incorporated under section 4.1 of the Joint PD+MR	
Documentation provided by project participant	
Revised joint PD-MR	
DOE assessment	Date: 22/02/2023
1. The seven-year generation value has now been incorporated under the section 5.2 of the submitted revised PD + MR which is further cross checked and found acceptable. 2. Estimated baseline emissions sample calculation has now been incorporated under the submitted revised monitoring report, section 4.1. Hence, finding (CL – 01) has now been closed.	

CL ID	02	Section no.	3.1	Date: 05/01/2023
Description of CL				
Under Section 1.9 of submitted Joint PD Monitoring Report • Information mentioned as “Estimated Lifetime of the project instance 1 technology equipment” and lifetime of the project considered as “25 years”. Therefore, with reference to VCS Joint PD monitoring template filling instructions, V4.1. PP shall clarify how this information relevant to current section. • The project lifetime has been considered as “25 years” however same has been considered under submitted IRR sheet as 20 years & other sections of the report. Therefore, PP’s requested to clarify the same along with evidences. 1) Further, PP’s requested to submit the solar module manufacturer data sheet which are installed at project site.				
Project participant response				Date: 18/01/2023
1. The information relating to estimated lifetime of the project activity is not relevant to the section 1.9 of the Joint PD-MR in line with the template filling instructions and therefore the section is modified to remove the statement regarding the lifetime of the project activity. 2. The life time of the project activity considered as 25 years based on the warranty on the Solar PV module and section 1.11 of the Joint PD-MR has been corrected to specify the lifetime of the project activity as 25 years.				
Documentation provided by project participant				
1. Revised joint PD-MR 2. Module warranty certificate				

DOE assessment	Date: 31/01/2023
1. Information relating estimated lifetime of the project activity has now been removed from the section 1.9 of the Joint PD-MR which is further checked by the assessment team and found acceptable. Therefore, this observation is closed now. 2. The life time of the project activity considered as 25 years consistent with submitted warranty Certificate of Solar PV Module and information from section 1.11 of the joint PD-MR has now been removed which is further checked by the assessment team and found acceptable. Therefore, this observation is closed now.	
Hence, Finding (CL- 02) is now being closed.	

CL ID	03	Section no.	3.1	Date: 05/01/2023
Description of CL				
Under Section 1.11 of submitted Joint PD Monitoring Report, information encapsulated under footnote '5' that the project is grouped project activity whose commissioning completed on 12/03/2021 which is observed to be inconsistent with information provided under section 1.2. & 1.4. Therefore, PP's requested to clarify the same along evidences.				
Project participant response				Date: 18/01/2023
1. The project activity is not a grouped project activity and therefore the footnote 5 under section 1.10 is modified 2. The start date of the project activity/crediting period is the date of commissioning of project activity and certificate of commissioning is submitted to establish the start date.				
Documentation provided by project participant				
1. Revised joint PD-MR 2. Commissioning certificate				
DOE assessment				Date: 31/01/2023
The incorporated information under footnote '5' has now been revised and now consistent with section 1.2 & 1.4 that, project is not a "grouped project activity". Further the information and start date of the project activity is checked & verified by the assessment team & found acceptable.				
Hence, finding (CL-03) has now being closed.				

CL ID	04	Section no.	3.1	Date: 05/01/2023
Description of CL				
Under Section 1.12 of submitted Joint PD Monitoring Report, "Project Location", the physical location mentioned under the section observed to be inconsistent while checking the coordinate. Therefore, PP's further requested to clarify the same along with evidences.				
Project participant response				Date: 18/01/2023
The project location has been modified to include the name of the 'union' (administrative division of Bangladesh) in which the project activity is being located.				
Documentation provided by project participant				
Revised joint PD-MR				

DOE assessment	Date: 31/01/2023
The project location has now been modified and found consistent with the project plant site. However, maps attached are observed to be not visible & clear and also the position of arrow direction is also not correct. Therefore, this observation is currently open. Hence, finding (CL-04) is currently open	
Project participant response	Date: 15/02/2023
1. Visible map is included under section 1.12 of the joint PD MR 2. The highlighted area appearing as arrow in the map is actually the demarcation of the project boundary.	
Documentation provided by project participant	
Revised joint PD-MR	
DOE assessment	Date: 22/02/2023
Enlarge and visible map has now been incorporated under the section 1.12 of the revised PD- MR which is further cross checked by the assessment team and found acceptable. Therefore, this observation is close now. Hence, finding (CL – 04) is closed now.	

CL ID	05	Section no.	3.4.1	Date: 05/01/2023
Description of CL				
Under Section 3.1 of submitted Joint PD Monitoring Report, “Tittle and Reference of Methodology”, the reference of Tool 32 “Positive List’s of Technologies” has been given however there is no application of the referred tool for current project activity has been observed under the report. Therefore, PP has to clarify the relevance of the Tool under current project activity.				
Project participant response				Date: 18/01/2023
Tool 32 is no longer applicable to the project activity and therefore not applied in the project activity for demonstration of additionality and hence the reference of Tool 32 is removed from section 3.1. of the Joint PD Monitoring Report.				
Documentation provided by project participant				
DOE assessment				Date: 31/01/2023
The reference of tool 32 has now been removed from the revised Joint PD Monitoring Report and justification provided by PP for non applicability of the tool for current project activity is checked & verified by the assessment team and found acceptable. Hence, Finding (CL- 05) is closed now.				

CL ID	06	Section no.	3.4.5	Date: 05/01/2023
Description of CL				
Under Section 3.5 of submitted Joint PD Monitoring Report, “Additionality”, covering sub heading “Common Practice Analysis”, the start date of the project activity has been considered as “Cut-off date of investment analysis”. Therefore, PP’s further requested to clarify the cut-off date of investment analysis along with supportive evidences consistent with CDM definition and incorporate the relevant date under section as well. - There is no reference to common practice analysis tool, version 3.1 has been provided under section 3.1 however same has been referred under section 3.5. PP’s has to clarify the same.				
Project participant response				Date: 18/01/2023
- Date of EPC contract between M/s Spectra Solar Park Limited and Sungrow Power PTE Ltd. (dated 12/12/2019) resembling project participant commitment for expenditure is considered as the start date for undertaking of the common practice analysis and is included under section 3.5 of the Joint PD Monitoring Report. 2) Reference of the Tools for common practice analysis is included under section 3.1 of the Joint PD Monitoring Report				
Documentation provided by project participant				
a. Common Practice Analysis Sheet b. Turnkey EPC contract (extract) c. List of Installed Grid Connected Power Plant in Bangladesh -Bangladesh Power Development Board (Doc Date – 07/09/2021) d. Bangladesh Power Development Board – Annual Report 2018-2019 and 2019-2020				
DOE assessment				Date: 31/01/2023
- The start date for undertaking of the common practice analysis has now been incorporated under section 3.3 of the Joint PD Monitoring Report which is further checked with submitted EPC contract between M/s Spectra Park Limited and Sung-row Power PTE Ltd. (dated 12/12/2019). Therefore, this observation is closed now. - The reference of “common practice analysis” is included under section 3.1 of the Joint PD Monitoring Report which is further checked with submitted Common Practice Analysis Sheet along with above mentioned documents and found acceptable. However, the total no of projects considered under the submitted common practice analysis sheet observed to be inconsistent with revised Joint PD Monitoring Report. Also, from the above mentioned document, I and (d) observed to missing under submitted set od documents. Therefore, this observation is currently open.				
Hence, Finding (CL- 06) is currently open.				
Project participant response				Date: 15/02/2023
The total number of projects considered under common practice analysis is modified under section 3.5 of the Joint MR+PD				
Documentation provided by project participant				
a. Revised Joint PD+MR b. List of Installed Grid Connected Power Plant in Bangladesh -Bangladesh Power Development Board (Doc Date – 07/09/2021) c. Bangladesh Power Development Board – Annual Report 2018-2019 and 2019-2020				

DOE assessment	Date: 22/02/2023
Document having serial no (b), dated 07/09/2021 under the PPs response, still observed to be missing from the submitted set of the documents. Therefore, PP shall be requested to cross check and furnish the same.	
Hence, finding (CL – 06) is currently open	
Project participant response	Date: 23/02/2023
The reference document is submitted (web link of the said document provided - https://bpdh.portal.gov.bd/sites/default/files/files/bpdh.portal.gov.bd/page/f883c9a0_a2dc_4f49_81de_f35d31650d48/2021-09-07-14-13-b3c60bbceb75b9ce13ee40effd3b64ac.pdf)	
Documentation provided by project participant	
List of Power plant along with commissioning date	
DOE assessment	Date: 07/03/2023
The document (b.) pertaining details of the power plant along with commissioning date has now been furnished by the project proponent which is further checked and verified by the assessment team and found acceptable.	
Hence, this observation (CL – 06) is closed now.	

CL ID	07	Section no.	3.4.6	Date: 05/01/2023
Description of CL				
Under Section 4.1 of submitted Joint PD Monitoring Report, "Baseline Emissions"				
- For the calculation of Operating Margin, the data for the year 2017-18, 2018-19, 2019-20 has been considered respectively. However, the value calculated for the similar projects registered under VCS registry is inconsistent. - And also, the generation value taken from "Annual Reports of Bangladesh Power Development Board" for the respective years are observed to be different mentioned under Ex-ante Estimated ER sheet.				
Therefore, PP's requested to cross check and clarify the same along with supportive				
Project participant response				Date: 18/01/2023
- The value of Net Electricity Generation, Plant load factor and Operation efficiency of the power plant used for calculation of the operating margin is obtained from the annual report published by Bangladesh Power Development Board - Bangladesh Power Development Board - Annual Report 2017-2018¹¹ (Chapter 4, pg 50) - Bangladesh Power Development Board - Annual Report 2018-2019¹² (chapter 4, pg 50) and - Bangladesh Power Development Board - Annual Report 2019-2020¹³ (chapter 4, pg 56)				

¹¹ https://bpdh.portal.gov.bd/sites/default/files/files/bpdh.portal.gov.bd/annual_reports/7b792f67_bf50_4b3d_9bef_8f9b568005c9/2022-10-18-05-39-77e3d834e198d21cdfb8d05d5c6c63b0.pdf

¹² https://bpdh.portal.gov.bd/sites/default/files/files/bpdh.portal.gov.bd/annual_reports/7b792f67_bf50_4b3d_9bef_8f9b568005c9/2022-10-18-05-38-b38d9a259adcd65ecb411b5ba4637b76.pdf

¹³ https://bpdh.portal.gov.bd/sites/default/files/files/bpdh.portal.gov.bd/annual_reports/7b792f67_bf50_4b3d_9bef_8f9b568005c9/2022-10-18-05-37-2ae06b4716a042c4b39d5cb54ef20ff4.pdf

o In addition, the emission factor for respective fossil fuel was obtained from Annex IV, CDM Baseline, Department of Environment, Bangladesh (http://www.doe.gov.bd/site/publications/61a68812-014c-405b-8218-eb1c87b2ec97/Clean-Development-Mechanism-CDM-Baseline-Power-and-Energy-Sector) The only registered project from Bangladesh - https://registry.verra.org/app/projectDetail/VCS/2263 " has considered the grid emission factor from "Clean Development Mechanism (CDM) Baseline, Power and Energy Sector", published by Department of Environment, Ministry of Environment and Forest, Bangladesh in 2012. The same value is being considered for estimation of emission reduction in the revised version of joint PD+MR. . The ER estimation in the joint PD+MR is revised based on the published value combine margin as is used in the project from Bangladesh registered with VCS registry. The value of generation is obtained from respective years annual report published by Bangladesh Power Development Board. Since default value of emission factor as published in CDM Baseline, Department of Environment is considered for estimation of emission reduction, the value of generation is not referred to in the revised PD+MR.	
Documentation provided by project participant	
DOE assessment	Date: 31/12/2023
- Justification provided by the project proponent is un-acceptable as data used for the calculation of operating margin under the referred registered project in Bangladesh is recent compared to 2009-10, 2010-11, 2011-12 is used for current project activity. Therefore, PP's further request to clarify with valid supportive. Observation is currently open. - The above mentioned document/report of default value of emission factor is considered from published CDM baseline, by Department of Environment for estimation of emission reduction is observed to missing from the submitted set of documents. Therefore, this observation is currently open. Hence, Finding (CL- 07) is currently open	
Project participant response	Date: 15/02/2023
1. Department of Environment, Ministry of Environment and Forest, Government of the People's Republic of Bangladesh had under CDM Capacity Building and Baseline Development Project published the grid emission factor for Bangladesh Grid in December 2012 3) http://www.doe.gov.bd/site/publications/61a68812-014c-405b-8218-eb1c87b2ec97/Clean-Development-Mechanism-CDM-Baseline-Power-and-Energy-Sector The combined margin estimated under the said publication was based on related information for 2009-10, 2012-11 and 2011-12 and considering 50% weightage for build and operating margin. The estimated combined margin emission factor in accordance to the said publication was 0.665 tCO₂/MWh. Estimated combined margin emission factor considering 75% operating margin and 25% build margin was 0.6605 tCO₂/MWh. In furtherance Department of Environment in 19/08/2013 re-notified the grid emission factor for the country as 0.67 tCO₂/MWh http://www.doe.gov.bd/site/notices/059ddf35-53d3-49a7-8ce6-175320cd59f1/Grid-Emission-Factor(GEF)-of-Bangladesh . The registered project from Bangladesh referred to the emission factor of 0.67 tCO₂/MWh.	

Since the last published combined margin emission factor data was of 2013, grid emission factor is calculated based on the most recent installed capacity and generation for the year 2017-18, 2018-19 and 2019-20 in line with Tool 07: Tool to calculate the emission factor for an electricity system, Version 07. The estimated combined margin emission factor based on the most recent information is 0.6585 tCO₂/MWh. Since the estimated value of grid emission factor is conservative than the combined margin emission factor published in 2012 and 2013, the estimated value of grid emission factor is considered for estimation of emission reduction. 2. The referred document is submitted.	
Documentation provided by project participant	
1. Bangladesh Power Development Board - Annual Report 2017-2018 (Chapter 4, pg 50) 2. Bangladesh Power Development Board - Annual Report 2018-2019 (chapter 4, pg 50) and 3. Bangladesh Power Development Board - Annual Report 2019-2020 (chapter 4, pg 56) 4. CDM baseline – 2012 5. Notification for grid emission factor 2013 6. Grid emission factor estimation sheet.	
DOE assessment	Date: 22/02/2023
The documents pertaining to annual generation for the year 2017-2018, 2018-2019 and 2019-2020 has now been received along with CDM baseline 2012 & Notification for grid emission factor 2013. However, under the above justification it's mentioned that, the single registered project in Bangladesh used, old 2013 emission factor i.e. 0.67 tCO₂/MWh which is further during the assessment observed to be inconsistent, as from the referred attached weblink it's observed that, for the registered project, the latest vintage year (2016-17, 2017-18 & 2018-19) generation data has been used and emission factor of 0.62 tCO₂/MWh incorporated for the calculation of emission factor. As 0.62 tCO₂/MWh is more conservative than the applied emission factor for the project activity i. e. 0.65 tCO₂/MWh. Therefore, PP shall be requested to cross check and clarify the same. Hence, finding (CL – 07) is currently open.	
Project participant response	Date: 23/02/2023
The grid emission factor has been revised (currently estimated as including import) based on the vintage year 2017-18, 2018-19 and 2019-2020 and revised grid emission of 0.6145 tCO₂/MWh is now considered for estimation of baseline emission and emission reduction. The estimated grid emission factor is lower than 0.6245 tCO₂/MWh considered in the registered joint PD.	
Documentation provided by project participant	
Revised Grid Emission Factor estimation sheet Revised Joint PD+MR	
DOE assessment	Date: 07/03/2023
The grid emission factor has now been modified with value consistent with vintage year generations 2017-18, 2018-19 and 2019-20 (including import) values which further cross checked and found acceptable to verification team and therefore, this observation is closed now. Hence, finding (CL -07) is closed now.	

Table 2. CAR from this verification

CAR ID	01	Section no.	Template	Date: 05/01/2023
Description of CAR				
Under cover page & template of submitted Joint PD Monitoring Report,				
- With reference to template filling instructions for Joint PD Monitoring Report v4.1, PP shall cross check the table format and revise under the cover page & throughout the report. - Cross check Footnote '5', "Statement" whose information mentioned under footnote observed to missing or unmarked, PP shall cross check and revise the same. - Under section 1.9, the scale of the project observed to be inconsistent with reference to template filling instructions, PP shall cross check and revise the same.				
Project participant response				Date: 18/01/2023
- The table format of the cover page of the Joint PD Monitoring Report is modified in accordance to the template filling instructions for Joint PD Monitoring Report. - Footnote 5 have been modified - The scale of the project has been revised under section 1.10 in line with the template filling instructions.				
Documentation provided by project participant				
Revised joint PD-MR				
DOE assessment				Date: 31/01/2023
- The font size of the revised Joint PD Monitoring Report, for cover page has now been corrected along with document version and date of issuance of the document. - Footnote 5 has now been revised and incorporates the information regarding start date of the crediting period. - The scale of the project has now been revised with reference to template filling instructions, version 4.1				
Hence, Finding (CAR – 01) is closed now.				

CAR ID	02	Section no.	3.1	Date: 05/01/2023
Description of CAR				
Under section 1.3 of submitted Joint PD Monitoring Report, the reference of VCS standard version 4.2 has been taken which is currently updated with latest version effective from 21 Dec 2022. Therefore, PP's requested to cross check the updated standard and incorporate the same for current project activity.				
Project participant response				Date: 18/01/2023
The version of the VCS standard is modified to "version 4.4." and the section 1.3 of the joint PD-MR is modified in line with the updated standard.				
Documentation provided by project participant				
Revised joint PD-MR				
DOE assessment				Date: 31/01/2023

The updated version of VCS standard version 4.4, has now been applied under the revised Joint PD Monitoring Report which is further checked and verified by the assessment team & found acceptable.

Hence, Finding(CAR – 02) has now been closed.

CAR ID	03	Section no.	3.1	Date: 05/01/2023
Description of CAR				
Under section 1.11 of submitted Joint PD Monitoring Report, For the table “technical specification” PP shall incorporate the following details:-				
• Make and wattage of modules installed • No. of installed solar modules with reference to each wattage type. • Total no of Inverter, make of inverter and capacity of each type of inverter installed. • Total no of transformers installed along with their capacity. • Monitoring Equipment along with substation detail where it’s installed.				
Further, PP’s requested to submit the Electrical Single Line Diagram and Solar Module layout of the project activity.				
Project participant response				Date: 18/01/2023
• The make (SUNTECH) and wattage (330Wp) of the modules installed is included under section 1.11. • Details pertaining to number of solar panel (137,520) of specific capacity (330 Wp of solar module) is included in the revised PD+MR. • Information pertaining to total no of Inverter (12 nos), make of inverter (SUNGROW) and capacity (rated output power - 3125 kW) of each type of inverter installed is included in the revised PD+MR. • Information pertaining total no of transformers (06 nos),installed along with their capacity (6.25 MVA) is included in the revised joint PD+MR. • Location of the Meters (both main meter and backup meter) along with name of the substation is included in the revised joint PD+MR. • SLD and module layout of the plant is submitted				
Documentation provided by project participant				
1. Revised joint PD+MR 2. Single Line diagram 3. Solar module layout submitted.				
DOE assessment				Date: 31/01/2023

- Make and wattage of installed solar modules has now been incorporated under the section 1.11 of revised Joint PD Monitoring report.
- Number of solar panels has now been incorporated under the table constitutes total DC capacity 45.381 MW_{DC} which is further checked with submitted Module layout and found inconsistencies with total numbers, DC and AC capacities of Modules. Therefore, PP shall cross check and revise the same.
- Total number of inverters along with their rated output capacity has now been incorporated under the table of section 1.11 of the revised report which is further checked with submitted Module layout and found inconsistencies in total numbers of inverters installed. PP shall cross check and revise the same
- Total number of transformers installed at project site has now been incorporated under the revised joint PD Monitoring Report which is further checked with submitted SLD and Module layout and found acceptable.
- Location of the meters (both checked and backup meter) along with name of the substation has now been incorporated under the table of section 1.11 of revised Joint PD Monitoring Report which is further checked with submitted SLD and Module layout and found acceptable

Hence, finding (CAR- 03) is currently open

Project participant response	Date: 15/02/2023
Revised module layout plan submitted that clearly outlines 137,520 number of solar module and 12 Nos of Inverter.	
Documentation provided by project participant	
Revised Module layout plan	
DOE assessment	Date: 22/02/2023
The revised module layout encapsulating total modules no's 137,520 having capacity of 330 Wp constituting total capacity 45.381 MW _{DC} along with additional details pertained to installed on grid inverters has now been submitted along with revised PD-MR which is further checked and found acceptable to the verification team.	
Hence, Finding (CAR – 03) has been closed now.	

CAR ID	04	Section no.	3.3.2	Date: 05/01/2023
Description of CAR				
Under section 2.2 of submitted Joint PD Monitoring Report, "Local Stakeholder Consultation", PP shall incorporate information regarding procedure & methods used for documenting the outcome of local stakeholder consultation, details of engagement of local stakeholders, mechanism of ongoing communication with local stakeholders etc. along with submission of following documents:				
• Grievance Register maintained at project site • Minutes signed during Local stakeholder consultation • Attendance sheet along with the participant stakeholder names • Public information shared for LSC to among local people etc. along with other supportive documents associated with Section				
Project participant response				Date: 18/01/2023

Stakeholders' consultations were carried out in two modes.	
1. In one mode ESIA consultant with support from representative of project proponent carried out wide stakeholders' consultation at 10 locations involving wider group of stakeholders on 13th,14th and 15th February 2019. 2. In the 2nd mode project proponent carried out stakeholder's consultation meeting at the plant premise on 14th February 2019 involving representative from nearby villages.	
Documentation provided by project participant	
1. Grievance Register 2. Minutes signed during Local stakeholder consultation 3. Attendance sheet along with the participant stakeholder names 4. Public Notice for inviting local personnels for stakeholders consultation meeting. 5. Feedback forms 6. ESIA Report	
DOE assessment	Date: 31/01/2023
Section 2.2, of revised Joint PD, Monitoring Report, now incorporating the information regarding procedure & methods used for documenting the outcome of local stakeholder consultation, details of engagement of local stakeholders, mechanism of ongoing communication with local stakeholders etc which is further verified by the submitted ESIA report, Public Notices, attendance sheet etc and found acceptable.	
Hence, finding (CAR – 04) is closed now.	

CAR ID	05	Section no.	3.4.1 & 3.4.2	Date: 05/01/2023
Description of CAR				
1. Under section 3.1 of submitted Joint PD Monitoring Report, "Title and Reference of Methodology" • The applied version of Methodology ACM0002 has now been updated, PP shall cross check and revise the same. • Referred tools for the current project activity shall not be incorporated under the section, PP shall cross check the applicable tools and incorporate reference of tools accordingly along with associated web link under footnote. 2. Under section 3.2 of submitted Joint PD Monitoring Report, "Applicability of Methodology", PP shall cross check and incorporate the applicability conditions of latest version of applied methodology and tools as well.				
Project participant response				Date: 18/01/2023
1. The applied version of the Methodology ACM0002 has now been revised under section 3.1. 2. Title and reference of the applied tools included. 3. Applicability section of the methodology and the tools is revised.				
Documentation provided by project participant				
Revised version of Joint PD + MR				
DOE assessment				Date: 31/01/1993

1. Under section 3.1 of revised Joint PD Monitoring Report
 - The version of applied methodology has now been changed which is further checked and found acceptable.
 - The referred applicable tools to current project activity has now been updated along with their weblink incorporation under the footnote.
2. The applicability conditions of the revised methodology and referenced tools has now been updated under the section 3.2 of the revised Joint PD , Monitoring report which is further check and found acceptable by the assessment team.

Hence, finding (CAR – 05) is closed now

CAR ID	06	Section no.	3.4.5	Date: 05/01/2023
Description of CAR				
Under section 3.5 of submitted Joint PD Monitoring Report, “Additionality” • Version no of Tool 27, “Investment Analysis” has now been updated, therefore, pp shall be requested to cross check and revise the version & applied value of benchmark used of Bangladesh. • Value of Inflation Forecast observed to be inconsistent with attached weblink under the Benchmark table. PP shall cross check the value and revise the same. Under the submitted IRR sheet, tab “P & L” the residual value has not been considered under cash flow statement, PP shall cross check & revise the same and also under tab “Assumption” the source of data shall be furnish along with cross check document of applied value.				
Project participant response				Date: 18/01/2023
• Revised version of the tool 27 is applied and the benchmark has been revised accordingly. • Value of Inflation Forecast have been revised. • The residual value (10% of the capital cost) has already been considered for estimating of net Depreciable value by subtracting the salvage value from the gross depreciable value. Assumption and cross check data submitted.				
Documentation provided by project participant				
1. ADB report submitted to substantiate the inflation rate 2. IMF Inflation rate submitted to cross check the inflation forecast 3. DPR 4. EPC Contract (Extract) + LC copy 5. O&M cost 6. Contract extract for warranty of products 7. Loan document 8. Project proponent declaration of capital cost 9. Land acquisition details and single agreement of land purchase				
DOE assessment				Date: 31/01/2023

- Tool 27, Investment Analysis has now been updated with revised version throughout the report which is further checked and found acceptable. Therefore, this observation is closed now. - Value of Inflation forecast has now been revised along with their associated web link which is further checked and accepted by the assessment team. - With reference to clause no 7 & rationale of Tool 27 of investment analysis, version 12, the residual value should be reflected under "cash flow". Therefore, PP's further requested to cross check and revise the same under the submitted IRR sheet. Observation is currently open. Further, also requested to furnish the ADB report , IMF inflation rate documents (above mentioned) as observed to be missing from the submitted set of documents.	
Hence, finding (CAR – 06) is currently open	
Project participant response	Date: 15/02/2023
Residual value of plant and machinery has been included as a part of IRR calculation with current IRR modified to 10.82% from 10.41%.	
Documentation provided by project participant	
1. Revised IRR sheet 2. Revised Joint PD+MR 3. ADB report 4. IMF inflation document (the Excel sheet for IMF inflation can also be downloaded from https://www.imf.org/external/datamapper/PCPIPCH@WEO/BGD?zoom=BGD&highlight=BGD inflation forecasting for 10 years period from 2018 to 20	
DOE assessment	Date: 22/02/2023
Residual value under the revised IRR sheet tab "P & L" for cash flow statement, has now been incorporated and corresponding IRR value under the IRR sheet and in revised PD- MR has been updated which is further checked by the verification team and found acceptable.	
Hence, finding (CAR – 06) has been closed now.	

CAR ID	07	Section no.	3.4.6	Date: 05/01/2023
Description of CAR				
Under section 4.1 of submitted Joint PD Monitoring Report, "Baseline Emissions", with reference to template filling instructions, version 4.1, PP shall incorporate the formula quantification for estimated emission reductions along with sample calculation.				
Project participant response				Date: 18/01/2023
Section 4.1 have been revised in line with the template filling instructions, version 4.1				
Documentation provided by project participant				
Revised PD+MR				
DOE assessment				Date: 31/01/2023
Section 4.1 of revised Joint PD Monitoring report, missing the sample calculation of estimated emission reductions. Therefore, PP's further requested to cross check and update the same.				
Hence, Finding (CAR – 07) is currently open				
Project participant response				Date: 15/02/2023
Sample Calculation for estimation of baseline emission included under section 4.1 of the joint PD+MR				

Documentation provided by project participant	
DOE assessment	Date: 22/02/2023
Sample Calculation for the baseline emissions has now been encapsulated under section 4.1 of the revised PD- MR which is further cross checked by the assessment team and found consistent with Applied Methodology, ACM0002, version 21 requirements. Hence, finding (CAR – 07) has been closed now.	

CAR ID	08	Section no.	3.4.8	Date: 05/01/2023
Description of CAR				
Under section 5.2 of submitted Joint PD Monitoring Report, “Data & Parameters Monitored”, For Monitoring Equipment parameter, PP shall incorporate the details of monitoring equipment like Make, Accuracy class, Type, Serial No along with calibration frequency referred to any regulatory guidelines.				
Project participant response				Date: 18/01/2023
Details of the monitoring equipment in terms of equipment like Make, Accuracy class, Type, Serial No along with calibration frequency referred to in the PPA is included.				
Documentation provided by project participant				
1. Meter calibration certificate 2. Joint meter reading (outlining meter accuracy)				
DOE assessment				Date: 31/01/2023
Section 5.2 of revised Joint PD Monitoring Report, “Data & Parameters Monitored” has now been incorporated with monitoring equipment details like Make, Accuracy class, Type, Serial Number. However the calibration frequency of the meters (monitoring equipment) both main & check are not less than 180 days as per the PPA , clause 12.1 . Therefore, PP further requested to furnish the calibration certificate of main and check meter for the period from 12/03/2021 to 01/01/2022. Observation is currently open. Hence, finding (CAR – 08) is currently open				
Project participant response				Date: 15/02/2023
In accordance to section 12.1 of the PPA the implementing company (Spectra Solar Park Ltd) needs to undertake inspection, testing, and calibration of the Metering System upon installation under observation of BPDB (Bangladesh Power Development Board). Thereafter, BPDB shall test the Metering System at intervals of not less than 180 (one hundred eighty) Days. In case during testing any of the component of the Metering System is found to be outside acceptable limits of accuracy or otherwise not functioning properly, SPECTRA Solar Park Ltd will require to repair, recalibrate or replace such component of the Metering System or the Back Up Metering System, as the case may be. BPDB however tests the meter on an annual basis and certificate of meter calibration dated 19/03/2022 by BPDB is enclosed. The meter was tested to be found within the accuracy class range				
Documentation provided by project participant				
Calibration certificate dated 19/03/2022				

DOE assessment	Date: 22/02/2023
The calibration frequency Details of the monitoring equipment has now been incorporated consistent with clause 12.1 of the power purchase agreement. However, the calibration certificate submitted, dated on 19/03/2022 which is lies outside the current monitoring period. (12/02/2021 to 01/01/2022) Therefore, PP's further requested to furnish the calibration certificates covering the entire monitoring period. Hence, finding (CAR – 08) is currently open.	
Project participant response	Date: 23/02/2023
The power purchase agreement refers to calibration of the energy meter based on BPDB discretion of the accuracy of the meter operation without any indication of the calibration frequency/schedule. In line with the Gazette Notification by Bangladesh Energy Regulatory Commission (BERC) dated 31 December, 2019 (http://berc.portal.gov.bd/sites/default/files/files/berc.portal.gov.bd/page/a250b6fc_8bcf_4c96_bb20_3c3de230467a/2020-03-01-17-04-6c286949cb014aac31754ffa3e86bac2.pdf), the calibration of metering system is required to be carried out at intervals after 5 years (section 13.6.10), along with periodical yearly testing of meter and associated equipment's. In line with the guidelines of BERC, the meter has been tested/ and the accuracy level is assessed to be within the specified limit 1. Inspection and test report of energy meter dated 09/02/2021 and 2. Inspection and test report of energy meter dated 19/03/2022 The current monitoring period (12/02/2021 to 01/01/2022) is within the testing/calibration period specified above and in line with the national regulations.	
Documentation provided by project participant	
1. Inspection and test report of energy meter dated 09/02/2021and 2. Inspection and test report of energy meter dated 19/03/2022 3. Revised Joint PD+MR 4. Gazette Notification by Bangladesh Energy Regulatory Commission (BERC) dated 31 December, 2019	
DOE assessment	Date: 07/03/2023
Inspection and test report of energy meter has now been verified, dated on 09/02/2021 along with the notification "Gazette Notification by Bangladesh Energy Regulatory Commission (BERC) dated 31 December, 2019" which is further, checked and found acceptable by the assessment team and Hence, this observation is closed now. Hence, finding (CAR – 08) is closed now.	

CAR ID	09	Section no.	4.2	Date: 05/01/2023
Description of CAR				
Under section 6 of submitted Joint PD Monitoring Report, "Data & Parameters Monitored" for achieved emission reduction and removals • PP shall submit the Joint Meter Report, Invoices & calibration report of the monitoring equipment for the period 12/03/2021 to 31/01/2022. And also incorporate the invoice values along with JMR under submitted ER sheet, tab "Ex-post" for cross check.				

Project proponent shall check the nomenclature/notation applied under “baseline emissions” calculation & attenuate with the standard.	
Project participant response	Date: 18/01/2023
- The invoice values along with JMR values of net electricity generation are incorporated in the, tab “Bill-invoice” for cross check. Copy of the JMR and Invoice for the period 12/03/2021 to 31/01/2022 submitted. - The nomenclature /notation applied under “baseline emissions” calculation are modified in line with the methodology.	
Documentation provided by project participant	
- Billing statement - Invoice - Daily Generation Report	
DOE assessment	Date: 31/01/2023
- The invoice values along with the JMR values of net electricity generation are incorporated in the revised ER sheet which is further checked and verified with submitted Billing statement and invoices and found acceptable. However, the generation for the March 2021 (full month) has been applied despite of start date of monitoring period (12/03/2021 – 31/01/2022). Therefore, PP’s further requested to cross check and revise the same. Observation is currently open. - The nomenclature /notation applied under “baseline emission” has now been corrected under revised Joint PD Monitoring Report which is checked and found acceptable.	
Hence, finding (CAR – 09) will be currently open	
Project participant response	Date: 15/02/2023
- The JMR record is from 12/03/2021 – 31/01/2022 - The monthly generation is cross checked with the daily generation report and cumulative daily generation estimated at 3870.0121 MWh is higher than the JMR reading of 3828.9168 MWh. Since the JMR reading is more conservative the said value is considered for estimation of emission reduction.	
Documentation provided by project participant	
Modified ER sheet with DGR value for March 2021	
DOE assessment	Date: 22/02/2023
Agreed from the justification, further requested to demonstrate the same (By considering both JMR and DGR generation) under the tab "Bill and Invoice" for more clarification. Also, apply apportion formula for considered JMR reading (from 12/03/2021 to 31/03/2021) applied for the start month.	
Hence, finding (CAR- 09) is currently open	
Project participant response	Date: 23/02/2023
The proposed project activity (solar power plant) is connected to a single energy meter that record the export and import of electricity generated from the project power plant and there is no other power plant connected to the said meter. The solar power plant started its commercial operation from 12/03/2021 and the JMR includes the reading of both dates i.e. 12/03/2021 and 01/04/2021 and the difference of reading of both dates has been used to calculate the net export after factoring in the import. Since there is not other power plant connected to the said meter therefore apportion principle is not applicable for the power import and export recorded for the said period	

Documentation provided by project participant	
1. Revised Joint PD+MR 2. JMR March 2021	
DOE assessment	Date: 07/03/2023
The justification provided by the project proponent along with revised emission reduction sheet, joint PD + MR along with DGR & JMR of March 2021 which is further crossed check by the assessment team and found acceptable.	
Hence, finding (CAR – 09) is closed now.	

Table 3. FAR from this verification

FAR ID	01	Section No.	E4.4	Date: 07/03/2023
Description of FAR				
During the next periodic verification, verifying VVB must performed the on-site visit to verify the project description/ design and implementation				
Project participant response				Date: DD/MM/YYYY
Documentation provided by project participant				
DOE assessment				Date: DD/MM/YYYY

APPENDIX 3: COMPETENCE OF TEAM MEMBER AND TECHNICAL REVIEWER

Verification Team Member: -

No	Role	Type of resource	Last name	First name	Affiliation (e.g. name of central or other office of DOE or outsourced entity)	Involvement in			
						Desk review	On-site inspection	Interview(s)	Verification findings
1.	Lead Auditor/ Technical Expert	OR	Singh	Jitendra Mohan	TQC- Outsourced entity	Yes	Yes	Yes	Yes

Technical Reviewer and Approver of the Verification and Certification Report: -

No.	Role	Type of resource	Last name	First name	Affiliation (e.g., name of central or other office of DOE or outsourced entity)
1.	Technical reviewer (TR)	EI	Xue	Denny	Applus+ Certification
2.	Approver	IR	Calle de Miguel	Agustin	Applus+ Certification

Shorts CV of the Team:

- Mr. Jitendra Mohan Singh**, has done Advanced MSc in Sustainable Energy Systems and Management from International Institute of Management, University of Flensburg, Germany and B.Tech. in Agricultural Engineering from Allahabad University, India. He has more than 22 years of working experience in different organisations like IARI, IIT Delhi, ICAR, IRADe, CAPART, SMEC and Perenia Carbon and M B Power (Madhya Pradesh) Ltd. in the area of Agriculture, Energy & Environment and Climate Change. He also worked on contract basis (adhoc) as a RIT expert in UNFCCC from 2010 to 2013. Currently, he is associated with True Quality Certifications Private Limited and is empanelled with Applus+ Certification to carry out validation and verification related to GHG reductions projects. Mr. Jitendra Mohan Singh is based in Ghaziabad (Uttar Pradesh), India.
- Mr. Denny Xue** (Master's Degree in Environmental Engineering, Bachelor's Degree in Thermal Engineering) is an Auditor appointed by Applus+ LGAI for the GHG project assessment, auditing and technical review. He has more than 6 years of work experience in

CDM/GS4GG/VCS project assessment and technical review with Applus+. Before he joined Applus+ LGAI, he has been working for Shanghai Chuanji Investment and Management which is a CDM consultancy company as a project manager for CDM project development. Mr. Denny Xue is based in Shanghai, China.

APPENDIX 4: ABBREVIATIONS

Abbreviations	Full texts
CAR	Corrective Action Request
CDM	Clean Development Mechanism
CER	Certified Emission Reduction(s)
CL	Clarification request
CMS	Central Monitoring system
CO ₂	Carbon dioxide
CO ₂ e	Carbon dioxide equivalent
DNA	Designated National Authority
DOE	Designated Operational Entity
DR	Document Review
EF	Emission Factor
EIA	Environmental Impact Assessment
ER	Emission Reductions
FAR	Forward Action Request
GHG	Greenhouse gas(es)
OM	Operating Margin
GWP	Global Warming potential
VVB	Validation/Verification Body

APPENDIX 5: CALIBRATION DETAILS

As per the monitoring plan the meter calibration frequency is once in 5 years (as per the BERC 2019, guidelines for meter testing and calibration) thus no delayed calibration observed during this monitoring period.

S.I No.	Meter Make	Class	Meter Serial Number	Meter Initial Calibration Date	Valid Upto
1	Landis+Gyr E650	0.2s	42334949	09-February- 2021	08-February- 2026
2	Landis+Gyr E650	0.2s	42334947	09-Feb-2021	08-February- 2026